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PTB: A deep reinforcement learning method for flexible logistics service combination problem with spatial-temporal constraint

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ABSTRACT

The Fourth Party Logistics Service platform aims to provide customers with one-stop logistics services by combining logistics services from different logistics service providers. However, it is an important challenge to quickly build the logistics service combination with the maximum overall profit under the premise of satisfying spatial-temporal constraints. The traditional heuristic algorithm has the problem of low computing efficiency and insufficient adaptability to dynamic environment and demand uncertainty when dealing with the problem of large-scale logistics service combination with spatial-temporal constraints and uncertainty of service. Therefore, we propose the Proximal policy optimization algorithm integrating the Transformer network and the Bayesian networks (PTB). Firstly, a variable length sequence module based on the greedy strategy is used in the environment to generate the status information of the order, to solve the problem that the traditional reinforcement learning environment cannot handle any number of service providers. Then, a policy network integrating the Transformer and Bayes network is used to further optimize the accuracy and reliability of scheduling decisions. Meanwhile the PPO algorithm is used to update the network and constantly optimize the scheduling decisions of agents. Finally, the experimental results on four sets of logistics order scenes show that PTB can effectively solve the logistics service combination problem with spatial-temporal constraints, and the results of most logistics service combination tasks are better than other baseline models, and in large-scale order scenes, it has far more computational efficiency than the traditional heuristic algorithm, and shows good generalization ability.

1. Introduction

The Fourth Party Logistics Service Platform (4PL) is an innovative logistics service model that provides customers with comprehensive logistics and supply chain management solutions by integrating and optimizing resources. Different from traditional logistics services, 4PL platform itself does not directly control logistics resources, but acts as a coordinator and integrator to effectively integrate the resources and services of third-party logistics service providers (3PL) through advanced technology and management methods to meet the individual needs of customers (Huang et al., 2019). In the process of customizing logistics solutions for customers, the core challenge faced by 4PL platform is how to reasonably organize and use 3PL resources and services to build a logistics service portfolio with the largest overall profit under the premise of satisfying the spatiotemporal constraints of a large number of sub-logistics orders.

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We have summarized this core challenge as the Flexible Logistics Service Combination Problem with Spatial-temporal constraints (FLSCPS), which is important not only for improving logistics efficiency, reducing costs, enhancing customer satisfaction, and supporting sustainable development, but also for its key role in enabling efficient and flexible supply chain management. Addressing FLSCPS, ensuring optimal use of resources and improving the quality of decision making is essential to enhance the adaptability and flexibility of the entire logistics system. Although existing studies have discussed FLSCPS to a certain extent, for example, researchers have devoted themselves to improving the quality of logistics services and optimizing transportation routes (Zhang and Pan, 2023; Liu et al., 2024), the spatial-temporal constraints of logistics service combination are still not fully considered. This neglect can lead to significant challenges for 4PL platforms in servicing vehicle scheduling and service combination, as they often struggle to meet order demand across all regions and time frames. In addition, most solutions still rely on the use of traditional solvers or heuristic algorithms (Liu and Wang, 2015; Yang et al., 2014), and lack of targeted and efficient algorithms for the flexible logistics service combination problem with spatial-temporal constraints. The existing algorithms generally have the problem of insufficient generalization ability (Wang and Yu, 2010), which is difficult to synchronize and optimize multiple logistics data sets of different sizes, thus limiting their wide applicability and efficiency in practical applications. Therefore, we need to explore how to combine the goals of FLSCPS with the requirements of the industry to provide more accurate and efficient service configuration for the logistics industry, thus enhancing the adaptability and flexibility of 4PL. Fig. 1 further introduces the flexible logistics service combination problem with spatial-temporal constraints.

Fig. 1 shows the spatial-temporal constraints followed by the two logistics transportation schemes (Plan1 and Plan2) in the transportation process. In Fig. 1(a), which shows the spatial constraints of logistics services, we divide all third-party logistics service providers into three groups according to the type of service they provide. collection, trunk, and distribution. For the logistics service demand in the same region, there can be multiple logistics service providers to provide related logistics services, so for the logistics service demand of a batch of orders, it is necessary to coordinate the three services of collection, trunk line and distribution, and select the best provider in the corresponding service provider set, so as to maximize the overall transportation profit. And the condition of the departure of the service vehicle is that the load must not exceed the maximum load of the vehicle or the vehicle has been loaded. Plan1 and Plan2 show two different shipping scenes that can occur for a given order, involving a total of seven shipping nodes. Fig. 1(b) shows information about time constraints, such as the fact that collection service vehicles can only be transported between 9 a.m. and 11 a.m. and 3 p.m. and 5 p.m., trunk service vehicles need to leave after 8 a.m., and distribution service vehicles need to leave after 10 a.m. Fig. 1(c) shows the change in the profit of the two schemes as the transit time changes. As can be seen from Fig. 1(c), the profit decline of the transportation scheme increases significantly with the extension of transportation time, and long transportation time means more fuel consumption, higher labor costs, and potential vehicle maintenance costs. These increased costs will reduce the overall profit of the shipping task, while the extended time may also lead to a decrease in the quality of logistics services and increase the risk of damage to the goods, thus affecting customer satisfaction.

Logistics Service Combination Problem (LSCP) as an important research field, there have been many targeted studies. It mainly includes the dynamic mutual manufacturing and transportation routing Service selection of cloud manufacturing with multi-phase service demand matching (Aghili et al., 2021), the competitive dynamics view of diversified service portfolio of third-party logistics providers (Barker et al., 2021), and the use of DQN to solve the perceived service portfolio involving logistics QoS (Quality of Service) in cloud manufacturing (Liang et al., 2021), etc. Although these methods have proven their effectiveness in terms of the logistics service combination problem studied by each of them, the following challenges still exist for the flexible logistic service combination problem with spatial-temporal constraints.

Challenge 1. Large scale logistics order service combination needs to meet the need for fast resolution. In modern logistics, meeting the real-time computing needs of a large-scale logistics service combination is a key challenge, its importance due to the complexity and dynamic changes inherent in handling a large number of orders. Meanwhile logistics services involve multiple dimensions such as resource allocation and time management, each dimension is full of numerous variables and constraints, and in the face of large-scale logistics service combination problems, it is necessary to generate feasible solutions within a limited time. However, the existing solutions often adopt simple rules such as first-come-first-served or the earliest cut-off time priority, which cannot fully consider the dependency relationship between services and complex constraints, while the traditional heuristic algorithms often have

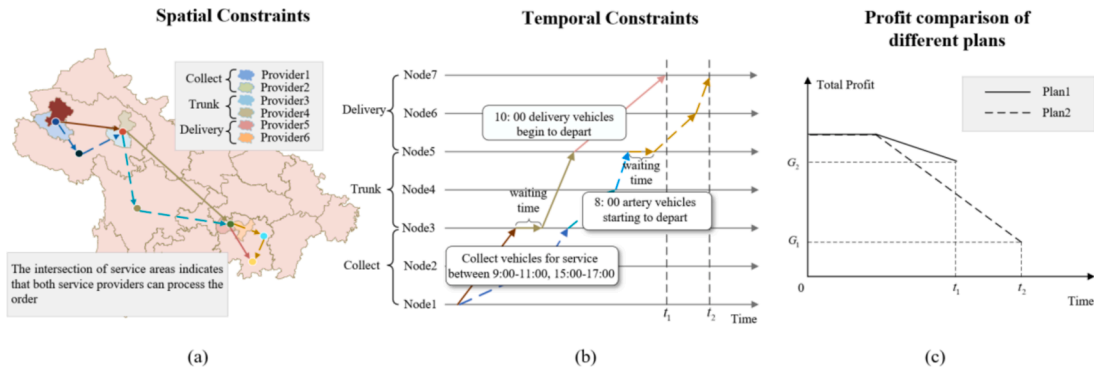


Fig. 1. Spatial-temporal constraints in the process of logistics order transportation.

the characteristics of low computational efficiency and weak generalization when facing such problems, so it is difficult to meet the business needs that require fast computation.

Challenge 2. Logistics service demand has spatial-temporal constraints and uncertainties. The time-space constraints of logistics services refer to that on the one hand, when customers place orders, they have clear requirements on the time and place of goods collection or distribution, and the space capacity of service vehicles limits the amount of goods they can handle. On the other hand, the logistics service provider can only handle the logistics service demand within its service area within the corresponding time window, and cannot handle the logistics service demand outside the service area. The uncertainty of logistics services stems from a variety of factors, such as start time, transportation time, sudden changes in customer demand, etc., which may lead to the adjustment or even interruption of the original plan. Therefore, in response to the challenges of spatial-temporal constraints and uncertainties, logistics services must adopt a highly flexible and adaptable combination strategy in order to quickly respond to changes and effectively deal with emergencies, which is crucial to improve the efficiency of logistics services and enhance customer satisfaction.

Challenge 3. Lack of network model to solve the flexible logistics service combination problem. The existing model is unable to solve the problem of logistics service combination of different scales. In practical applications, as the number of service providers changes, traditional methods cannot solve the order data with different logistics service providers. In order to ensure that the network training model can be applied to logistics service combination problems of different scales and structures, it is necessary to fully consider the generalization ability and adaptability of the model, so that it has good stability and reliability in practical application.

In response to the above challenges, this paper proposes a model PTB that uses deep reinforcement learning to solve FLSCPS, which aims to obtain a more accurate and efficient logistics service combination solution. This research goal is closely related to the industry requirements in the following aspects: First of all, the rapid solution of service portfolio solutions for large orders is the need of the logistics industry. In the logistics business, in the face of massive orders, how to deal with and distribute efficiently has become a major challenge for the industry. As an advanced processing method, the PTB model can process and respond to the changes of logistics orders in real time and improve the processing efficiency. By optimizing transportation processes and resource allocation, the PTB model can significantly improve the speed of decision making to meet customer needs for fast delivery. In a highly competitive market, this ability to serve high-volume orders quickly and accurately is key to increasing customer satisfaction and winning market share. Secondly, logistics cost optimization is one of the core goals pursued by logistics enterprises. Through the PTB model, enterprises can allocate resources more accurately, reduce unnecessary waste, and reduce operating costs, which not only improves the profitability of enterprises, but also helps to achieve the goal of green logistics and sustainable development. Moreover, the quality of decision is the direct reflection of logistics service ability. By optimizing the service combination solution, the PTB model not only improves the accuracy of decision making, but also enhances the enterprise's ability to adapt to market changes and control risks. In addition, with the increasing complexity and uncertainty of logistics environment, the dynamic adjustment ability of the PTB model is particularly important. It can adapt to market changes and order fluctuations, and maintain the continuity and stability of logistics services.

In short, the research and development of the PTB model respond to the needs of the logistics industry for intelligent, automated and efficient decision-making. By accomplishing these objectives, the PTB model is anticipated to assist logistics enterprises in enhancing operational efficiency, cutting costs, boosting customer satisfaction, and gaining a competitive edge in the intense market rivalry. Therefore, the results of this study not only have important theoretical value to the academic circle, but also have a profound impact on the practical application of logistics industry. The main contributions of this study can be summarized as follows.

- (1) To address the challenge of swiftly solving large-scale logistics order service combinations, we introduced a novel model called PTB, which is based on the proximal policy optimization algorithm. The PTB model not only delivers a satisfactory logistics service combination but also achieves optimal solutions swiftly, offering a significant edge to logistics companies that must rapidly adapt to market shifts.
- (2) Aiming at the problem that traditional reinforcement learning environment cannot process logistics order data with different number of service providers, we propose a variable length sequence module based on greedy strategy (GVLS). This module gives the model flexibility to process variable length order data series, and significantly improves the model's adaptability to changes in the actual logistics environment. This enhanced generalization is a key factor in ensuring that the model can be successfully implemented across a wide range of logistics practices.
- (3) To deal with the uncertainty of orders under spatial-temporal constraint, we build a policy network that integrates the Transformer and the Bayesian network. The policy network combines the sequence modeling capability of encoder-decoder architecture with the probabilistic reasoning mechanism of the Bayesian network, which can capture and deal with the complexity and uncertainty of logistics service more effectively, and improve the accuracy and reliability of logistics service combination solution.
- (4) Experiments conducted in four logistics order scenes show that the PTB model shows excellent performance in dealing with logistics service combination problems, and has obvious advantages over traditional methods. These experiments not only validate the efficiency of the PTB model in handling large-scale orders, but also highlight its excellent generalization ability in dealing with complex scenes.

The rest of the paper is organized as follows, Section II provides an overview of relevant mathematical models, heuristic algorithms, and deep reinforcement learning models within the field of logistics service combination. Section III mathematically models the logistics service combination problem with spatial-temporal constraints and constructs a Markov decision process for the problem. Section IV explains the GVLS module, the structure of the network model and the training process of DRL in detail. Section V verifies the effectiveness of the PTB model through various experiments, and compares and analyzes it with other baseline models. Section VI

introduces the management insights of the PTB mode. Section VII summarizes the whole paper and looks forward to future work.

2. Related work

2.1. Logistics service combination

At present, logistics service combination is the key factor restricting the efficient operation and customer satisfaction of logistics industry, aiming at this problem, academia and industry have carried out a lot of related research and practice exploration. The research on LSCP starts from solving the service quality of logistics and reducing the cost. In order to improve the accuracy and green level of logistics service selection, it provides decision support for supply chain management. By establishing game model (Song and Wang, 2020), grey relation analysis and order preference technology (Naseem et al., 2021), researchers help decision makers to select the best logistics service provider for e-commerce. Meanwhile researchers also focus on solving the problems of subjectivity and fuzziness in the selection of cold chain logistics service providers (Mavi et al., 2017; Wang et al., 2021), and propose a comprehensive decision-making method (Zhang et al., 2023), which uses fuzzy comprehensive evaluation method to generate supplier rankings and recommend the best suppliers. In addition, there are also relevant studies on the selection of green logistics service providers (Celik et al., 2016) and inventory control (Sun et al., 2022). These results provide an important reference for the selection of logistics service providers. Reducing operating and logistics costs is also an important part of improving the quality of logistics services. By establishing the LSS order insertion scheduling model (Liu et al., 2014) and proposing the optimal logistics service composition model (Aghamohammadzadeh et al., 2020), researchers comprehensively consider various factors in the scheduling process to achieve the minimization of cost benefits for logistics service providers.

In recent years, the field of logistics service has been widely concerned by various industries, and the improvement of logistics service level determines the competitiveness of the market. Vivaldini et al. discussed the application of fleet management technology in logistics service management (Vivaldini et al., 2012), and demonstrated that the combination of existing technology and fleet management system has good benefits through case studies. Qin et al. put forward an analytical model to investigate the interaction between sales mode selection and logistics service strategy (Qin et al., 2021), and found that supplier preference is consistent with the improvement of logistics service level. Gaudenzi et al. adopted a qualitative comparative analysis method to study logistics service quality and customer satisfaction from the perspective of supply chain quality (Gaudenzi et al., 2020). Meanwhile with the increasing use of third-party logistics (3PL) services by shippers worldwide, the need for careful selection of third-party logistics service providers is becoming increasingly important. Mes et al. proposed a synchronous planning algorithm (Mes and Iacob, 2016) to select the best combination of transportation modes for each transportation order through intermodal planning among logistics service providers. Raue et al. studied the impact of the interaction of different governance types on service providers in horizontal cooperation (Raue and Wieland, 2015). Bayazit et al. demonstrated how analytic hierarchy process can be used to help companies select the most capable third-party logistics service providers for aerospace companies (Bayazit and Karpak, 2013). Liu et al. established a two-stage order allocation model (Liu et al., 2018) that considered demand renewal and fair preference of functional logistics service providers, so as to achieve the goal of maximizing the profit of logistics service integrators. The optimization of logistics transportation plan also plays an important role in reducing the negative impact of urban logistics activities. Pimentel et al. established a mathematical model of urban logistics distribution service operation planning (Pimentel and Alvelos, 2018), thereby reducing the total service time of logistics distribution, improving the efficiency of passenger transport network and improving the living conditions of major urban centers. Although the above research progress has provided help to improve the quality of logistics, there is still a lack of research using algorithms to solve the problem. Du et al. (Du et al., 2023) and Li et al. (Li et al., 2018) both adopted optimization algorithms, using GSA and improved particle swarm optimization respectively to achieve service composition optimization. Wicaksono et al. (Wicaksono et al., 2019) proposed a mixed integer linear programming model to solve the problem of dynamic supplier and carrier selection. Meanwhile the spatiotemporal logistics network considers the dynamic changes of logistics activities in time and space, and can adjust logistics services in real time to meet market demand and environmental changes. The theory and practice of spatiotemporal logistics network provide a scientific basis and implementation framework for logistics service combination. Zhao et al. (Zhao et al., 2022) proposed a PL resource allocation method based on dynamic spatial-temporal knowledge graph to achieve reasonable and effective allocation of production logistics resources. Wang et al. (Wang et al., 2020) developed a three-component scheme framework to solve the problem of two-level collaborative multi-center vehicle routing based on state-spatiotemporal networks, thereby promoting collaboration and resource sharing in state-spatiotemporal networks. Wang et al. (Wang et al., 2021) built a dual-objective mixed integer programming model based on state-spatial-temporal network to meet customers' demand for important materials with the lowest cost and fastest emergency response speed and optimize vehicle routes under the condition of limited transportation resources. Wang et al. (Wang et al., 2020) used two-stage service to solve the problem of green logistics location path optimization considering ecological package. All these works focus on the comprehensive consideration of service quality, synergy and time, seeking the optimal solution and improving the efficiency of logistics services.

In the process of studying logistics service combination, the importance of logistics perception service cannot be ignored. Although there are relatively few researches on logistics awareness service, it is of great help to improve the reliability and efficiency of logistics service. Therefore, the inclusion of logistics awareness services in the research field is helpful to understand and solve the logistics service combination problem more comprehensively, so as to achieve the goal of cost minimization and operation optimization. Researchers have adopted different algorithms and methods to solve the logistics service combination problem, such as dynamic QoS-aware service combination based on social networks (Yu et al., 2014), mutual manufacturing and logistics service combination (Aghili et al., 2021), flexible QoS-aware service combination (Khanouche et al., 2020), and task demand optimization service combination

(Yuan et al., 2020). In addition, researchers have adopted some new concepts and techniques, such as Generalized Component Services (GCSs) (Wu et al., 2016), orthogonal genetic algorithms (OGA) (Bao et al., 2016), and time-object decomposition mechanisms based on time models (Wang et al., 2021). The introduction of these methods and technologies expands the range of service choices and provides better solutions. In addition, these algorithms emphasize the importance of multiple QoS attributes and predict the time-aware dynamic quality of Service (QoS) attributes of a service (Syu et al., 2018) and propose a candidate pruning algorithm (Chen et al., 2016) to help select the service that best fits the problem. These efforts can promote the development of logistics services and improve the efficiency and quality level of the entire industry.

2.2. Deep reinforcement learning

At present, some researches use heuristic algorithms and reference models to solve the logistics service combination problem. Although these methods provide some references in optimization, there are some problems such as poor data robustness, long training time and less than expected results. In order to solve these problems, some studies have begun to explore the use of deep reinforcement learning (DRL) methods to solve the logistics service composition problem. Researchers proposed DDPG (Liu et al., 2022) and PPDRL (Yi et al., 2022) models, trained agents to learn the optimal service combination solution, and further optimized the results by using policy-based training methods, thus improving the efficiency and accuracy of solving logistics service combination problems. And Order scheduling also aims to improve efficiency and accuracy by optimizing resource allocation and task scheduling. The research results of the order scheduling problem provide a useful reference for solving the logistics service combination problem. Kavuk (Kavuk et al., 2022) et al. used Deep Q-networks to learn the actions of warehouses and achieve the goal of ultra-fast service delivery. Tang et al. (Tang et al., 2019) proposed a cerebellar value network (CVNet) with a new distributed state representation layer to solve the carpool order scheduling problem. Holler (Holler et al., 2019) et al. used deep reinforcement learning to rationally allocate the resources of vehicles with multiple drivers through an end-to-end method to solve the relocation problem. Malus (Malus and Kozjek, 2020) et al. applied multi-agent reinforcement learning technology to solve the difficult problem of real-time order scheduling of autonomous mobile robot fleets. Tong (Tong et al., 2021) et al. combined reinforcement learning and combinatorial optimization to propose a systematic solution aimed at recommending the right driver for the order. The research results of these works have brought innovative ideas and methods to the field of order scheduling, which is of great significance for realizing efficient service and resource allocation. It is of positive significance to solve the logistics scheduling problem in different fields to promote the in-depth study of logistics service combination. These jobs cover a variety of scheduling scenes, including fresh delivery (Yu, 2022), cloud manufacturing (Wang et al., 2023), smart factories (Lei et al., 2023), and green logistics (Huang et al., 2022). By applying deep reinforcement learning methods, these algorithms can provide efficient scheduling schemes and optimize fleet management decisions. Combining solutions from these different areas helps further explore how different logistics services can be intelligently combined to reduce costs and meet user needs.

Both logistics service combination problem and job-shop scheduling problem belong to scheduling problem. Job-shop scheduling problem focuses on production activities in scheduling shop, while logistics service combination problem considers how to orderly combine different logistics services to achieve efficient logistics operation. What these two problems have in common is the need to find the optimal scheduling scheme to meet the complex business requirements. Therefore, similar optimization methods and techniques can be used to solve these problems. By effectively combining the job-shop scheduling problem and the logistics service combination problem, the operation efficiency of the whole logistics system can be better optimized, the resource utilization rate can be improved, the logistics cost can be reduced, and the customer satisfaction can be improved. At present, the research of JSP problem is also gradually developing. Yuan (Yuan et al., 2023) et al. modeled JSSP as Markov decision process (MDP) and designed a new state representation based on the state features of bidirectional scheduling to optimize the solution of job-shop scheduling problems. Although this approach can solve common JSP, it does not capture the dynamic environment changes of the shop floor well. Therefore, researchers proposed a DRL-based dynamic scheduling method for the dynamic changing manufacturing environment, such as random arrival of jobs, machine failure, uncertain processing time and other factors (Zhang et al., 2022; Wang et al., 2021), and constructed a dual-depth Q network (DDQN) (Chang et al., 2022). A composite scheduling action (Gui et al., 2023), which is synthesized by SDR (single scheduling rule) and continuous weight variables, is designed to adapt to the dynamic changes of the flexible job shop

Table 1
Comparison of relevant literature on logistics service combination problem.

Paper	Research problem	Objective	TS constraints	Solution algorithm
Yu, Y et al. (2014)	QoS	QoS cost	×	PartnerFirst algorithm
Mes et al. (2016)	ST	Transport cost	×	Synchromodal planning algorithm
Wicaksono, P.A et al. (2019)	DSSP	Total cost	×	A mixed integer linear programming model
Aghamohammadzadeh E et al. (2020)	CM	Manufacturing cost, Energy consumption	×	Multi-objective cloud-service composition optimisation algorithm
Wang, Y et al. (2021)	LD	Total logistics operating costs,	✓	Two-stage solution algorithm
Zhao, Z et al. (2022)	PL	Total travelling distance	✓	DSTKG
This paper	FLSCPS	Total profit	✓	PTB

*Notes: CM-cloud manufacture; ST-Syncho modal transport; PL- Production logistics; QoS- Quality of Service; DSSP- Dynamic supplier selection problem; LD-logistics delivery.

production environment, and finally achieve better scheduling results in the dynamic changes of the manufacturing environment.

Although these studies provide guidance for the logistics service combination problem, it is difficult to apply in practice to solve the flexible logistics service combination problem with spatial–temporal constraints because of its low computational efficiency and weak adaptability to dynamic environment and order uncertainty.

2.3. Summary of literature

In order to further illustrate the innovation of the model and problem in this paper, we compare some relevant literature on logistics service portfolio in Table 1, including the research problem, the objective, whether the time–space migration constraint (TS constraints) is included, and the solution algorithm. By observing the research content in Table 1, we can summarize three main points.

- (1) The current research is not enough to deal with the flexible logistics service combination with the spatial–temporal constraints. While much existing research has focused on improving the quality of logistics services and optimizing transportation routes, few studies have focused on innovative optimization strategies to provide optimal service provider selection for order transportation in the logistics supply chain.
- (2) The existing research has not fully considered the spatial–temporal constraints of logistics service combination. This neglect can lead to significant challenges for 4PL platforms in servicing vehicle scheduling and service combination, as they often struggle to meet order demand across all regions and time frames. Therefore, when solving the problem of logistics service mix, the constraints of the real world must be taken into account in order to achieve more accurate and efficient service configuration.
- (3) At present, most of the solutions still rely on the use of traditional solvers or heuristic algorithms, and lack of targeted and efficient algorithms for the flexible logistics service combination problem with spatial–temporal constraints. In addition, the existing algorithms generally have insufficient generalization ability, which is difficult to synchronize and optimize multiple logistics data sets of different sizes, thus limiting their wide applicability and efficiency in practical applications.

In the logistics industry, continuous improvement of management efficiency and decision-making quality is the eternal pursuit. However, relatively little research has been done on making decisions in FLSCPS. In view of this, aiming at the industry pain points faced by FLSCPS, it is particularly important to propose targeted reliable management technology, which needs more attention and exploration. The contribution of this paper is to propose a new PTB model, which shows excellent performance and generalization ability in dealing with logistics service combination problems. We verify the effectiveness of the PTB model in various scenes through experiments, and compare it with other baseline models. This research result provides new solutions and management measures for the logistics service combination problem.

3. Preliminaries

This section mainly introduces the problem definition, problem modeling and reinforcement learning Markov decision process, so as to establish the consensus and boundary of this problem, so as to ensure the correctness and repeatability of the analysis and results of the subsequent research.

3.1. Optimization objective

The FLSCPS is modeled as follows: m service providers $C = \{c_k\}, k \in (1, 2, \dots, m)$ transport n orders $O = \{o_i\}, i \in (1, 2, \dots, n)$ with multiple services, and each order o_i has a service set $D_i = \{d_i^j = (cs, ts, ds) | cs \vee ts \vee ds = 1\}, i = (1, 2, \dots, n), j = (1, 2, \dots, l)$ with l service requirements. cs, ts and ds each represents a Boolean variable with a value of 0 or 1, where cs indicates whether there is a collection service, ts indicates whether there is a trunk service, and ds indicates whether there is a delivery service. $d_i = (cs, ts, ds) | cs \vee ts \vee ds = 1$ indicates that the service set for order o_i is defined by a tuple containing three Boolean variables corresponding to collection, service, and trunk service. To form a valid service set, at least one of the three Boolean variables must be true. This means that the order must include at least one of the collection, service or trunk services. And define D_{c_k} as the set of services that service provider c_k can process, and $Ip(c_k, d_i^j)$ represents whether service provider c_k can process the j -th service requirement d_i^j of order o_i . The profit of service demand d_i^j is expressed as $p_i^j \in P_i$. In the transportation process of order o_i , once the transportation time exceeds the preset time threshold, its profit $\sum_{j=1}^l p_i^j$ will decrease exponentially with the accumulation of timeout time according to the attenuation coefficient α . The ultimate goal of solving FLSCPS is to maximize the overall profit of all service providers after completing a batch of orders together. Therefore, the objective function can be defined as.

$$\max \sum_{i=1}^n \sum_{j=1}^l p_i^j \times \alpha^{\tau_{d_i^j}^{timeout}} \quad (1)$$

where p_i^j represents the initial profit of the section j service of order o_i , α represents the attenuation coefficient of profit, and $\tau_{d_i^j}^{timeout}$ represents the timeout time of the section j service transportation of order o_i , which is defined as follows.

$$T_{d_i}^{timeout} = \begin{cases} 0, & \text{if } t - T_{o_i}^{latest} \leq 0 \\ t - T_{o_i}^{latest}, & \text{otherwise} \end{cases} \quad (2)$$

where t represents the current time step of the environment, and $T_{o_i}^{latest}$ represents the latest processing completion time of order o_i .

Service providers often face challenges posed by spatial-temporal conditions and order-specific service characteristics when performing shipping order tasks. Due to these restrictions, some orders may experience stagnation during shipping. So when solving FLSCPS, it involves a set of constraints. The FLSCPS constraints are as follows.

$$\sum_{k=1}^m No_{c_k}^t \leq m, No_{c_k}^t \in [0, 1] \quad (3)$$

$$Bool_{cs}^t = \begin{cases} 1, & \text{if } (9 \leq t \% 24 \leq 11) \vee (15 \leq t \% 24 \leq 17) \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

$$Bool_{ts}^t = \begin{cases} 1, & \text{if } t \% 24 \neq 8 \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

$$Bool_{ds}^t = \begin{cases} 1, & \text{if } t \% 24 \neq 6 \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

$$(T_{ts_{o_i}}^{begin} > T_{cs_{o_i}}^{end}) \wedge (T_{ds_{o_i}}^{begin} > T_{ts_{o_i}}^{end}), T_{cs_{o_i}}^{end} > 0, T_{ts_{o_i}}^{begin} > 0, T_{ds_{o_i}}^{begin} > 0 \quad (7)$$

$$Ip(c_k, d_i^j) = \begin{cases} 1, & \text{if } d_i^j \in Dc_{c_k} \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

$$Vd_t = \begin{cases} 1, & \text{if } cw_t \neq 0 \text{ or } cw_t \leq Lw_{max} \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

$$Os_{o_i}(t) = \begin{cases} 1, & \text{if } Co_{o_i}^t \text{ is true} \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

$$Nop_{o_i}(t) = Nop_{o_i}(t_{previous}) + \Delta Nop_{o_i}(t), Nop_{o_i}(t) \geq 0 \quad (11)$$

$$Nosp_{o_i}^t = 1 \wedge Ip(c_k, d_i^j) = 1 \quad (12)$$

$$n \geq 0, m \geq 0, l \geq 0, |Dc_{c_k}| \geq 0 \quad (13)$$

Formula (3) indicates that an order service can only be assigned once an hour, that is, the order service must stay at least one time unit with the assigned the service provider after each assignment. $No_{c_k}^t$ represents the number of services assigned to service provider c_k in time step t .

In formula (4), $Bool_{cs}^t$ means that the service provider limits the collection vehicle to receive new collection orders during two time periods of 9.00–11.00 and 15.00–17.00 each day. In formula (5), $Bool_{ts}^t$ means that the service provider allows the trunk vehicle to wait to receive logistics orders requiring trunk transport at times other than 8.00 am. In formula (6), $Bool_{ds}^t$ means that the service provider allows the delivery vehicle to wait at any time other than 6.00 AM to receive logistics orders that need to be delivered. The formula (4), (5) and (6) form the time constraints $Cons_{time} = (Bool_{cs}^t, Bool_{ts}^t, Bool_{ds}^t)$ of FLSCPS.

Formula (7) indicates that the transportation of each section of the logistics order cannot begin until the transportation of the previous section of the service is completed. Formula (8) defines that service provider c_k can only process order service requirements d_i^j within its own capabilities. Formula (9) defines that the condition for the departure of the service vehicle is that the load must not exceed the maximum load capacity of the vehicle or the goods not carried by the vehicle. Vd_t represents whether the service vehicle starts at time t , cw_t represents the load capacity of the service vehicle at time t , and Lw_{max} represents the maximum load capacity of the service vehicle. The formula (7)–(9) form the spatial constraints $Cons_{space} = (SSO, Vd_t, Op(c_k, d_i^j))$ of FLSCPS. Both $Cons_{time}$ and $Cons_{space}$ belong to the constraints of static information in the state space.

In formula (10), $Os_{o_i}(t)$ represents the state of the order o_i at time t . $Co_{o_i}^t$ is a Boolean variable that indicates whether the processing of order o_i is complete at time t , $O_{status}^i(t) = 1$ when $Co_{o_i}^t$ is true, and $O_{status}^i(t) = 0$ when $Co_{o_i}^t$ is false. Formula (11) defines that the number of picks for order o_i is time-dependent. $Nop_{o_i}(t)$ represents the number of picks for order o_i at time t , $t_{previous}$ is the previous point

in time, and $\Delta Nop_{o_i}(t)$ is the number of new picks for order o_i during the time interval from $t_{previous}$ to t . Formula (12) defines that at time t it is certain that an optimal service provider can be selected for order o_i and that this service provider can handle the j -th service requirement d_i^j of order o_i . $Nosp_{o_i}(t)$ represents the optimal number of service providers assigned for order o_i at time t . Formula (10)–(12) are constraints on dynamic information in the state space.

Formula (13) defines the range of the number of orders, the number of service providers, and the number of services ordered. $|Dc_{c_k}|$ represents the number of services that the service provider c_k can handle.

3.2. Markov decision process

The Markov decision process (MDP) corresponding to the logistics service combination problem can be constructed as a quadruple $\langle S, A, R, \alpha \rangle$, which is explained in detail below.

- (1) The state space S is composed of static information S_{static} and dynamic information $S_{dynamic}$, which can be represented as $S = (S_{static}, S_{dynamic})$. Where $S_{static} = \{Cons_{time}, Cons_{space}, R_{resource}, O_{attribute}^i\}$, $Cons_{time}$ represents the time constraints of the service, $Cons_{space}$ represents the space constraints, $R_{resource}$ represents the resource state, $O_{attribute}^i$ represents the order attribute, $O_{attribute}^i$ includes the generation time $T_{o_i}^{generate}$, volume V^{o_i} , service demand set D_i , deadline time $T_{o_i}^{deadline}$, and latest processing completion time $T_{o_i}^{latest}$, all of which are relatively stable and do not change with time. $S_{dynamic} = \{Os_{o_i}(t), Op(t), Nop_{o_i}(t), Nosp_{o_i}(t)\}$, where $Os_{o_i}(t)$ represents the status of the order o_i at time t , $Op(t)$ represents the processing progress of order at time t , $Op(t) = \frac{Nco(t)}{n}$, $0 < Op(t) \leq 1$, $Nco(t)$ represents the number of orders that have been processed and completed at time t . $Nop_{o_i}(t)$ represents the number of selections for order o_i at time t , $Nosp_{o_i}(t)$ represents the optimal service provider for order o_i at time t , and this information is updated over time. Static information will affect the update of dynamic information in the environment. Once the static information enters the network, it will not be changed, and the subsequent state information entering the network will be the updated dynamic information.
- (2) A represents the set of scheduling actions, $A = \{a_t^{ijk} | a_t^{ijk} \in \{0, 1\}, \forall t \in T, i \in (1, 2, \dots, n), j \in (1, 2, \dots, l), k \in (1, 2, \dots, m)\}$. $a_t^{ijk} = 1$ indicates that service provider c_k processes the j -th service demand of order o_i at time t . $a_t^{ijk} = 0$ indicates that the service provider c_k does not process the j -th service demand of order o_i at time t .
- (3) R represents the reward function in the interaction between the environment and the Agent. The service provider receives an immediate reward, denoted as r_t , as feedback each time it processes an order service. The optimization direction of MDP in FLSCPS is to maximize the cumulative reward that can be obtained in the decision-making process. Therefore, in the design of reward function R , the optimal scheduling strategy should enable MDP to obtain the maximum cumulative reward. The design of the reward r_t is that when an order is completed at time t , the reward obtained is the sum of the profit after time decay of the completed order, otherwise it is the inverse of the remaining uncompleted order number $n - Nco$, to correctly guide the agent towards the goal of quickly completing the order. The formula for the reward r_t is as follows.

$$r_t = \begin{cases} \sum_{i=1}^{Nco} \sum_{j=1}^l p_i^j \times \alpha^{\tau_{timeout}^{d_i^j}}, & \text{if } Oct_i \text{ is true} \\ -(n - Nco), & \text{otherwise} \end{cases} \quad (14)$$

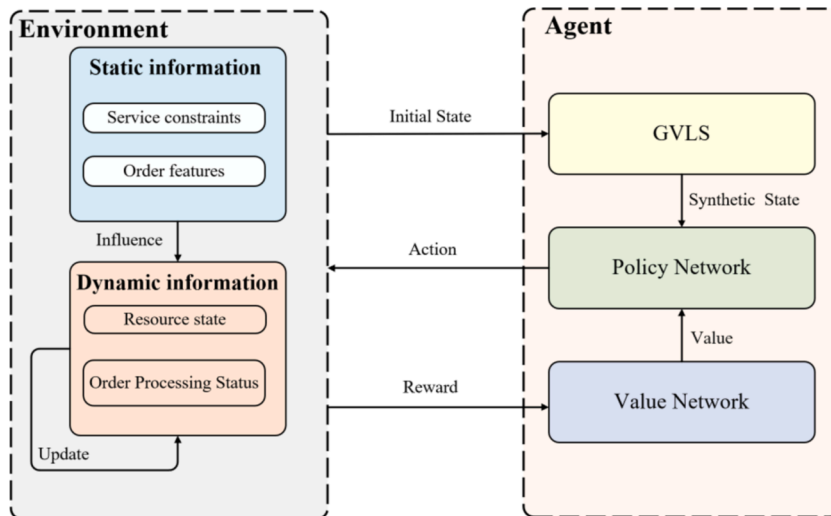


Fig. 2. Deep reinforcement learning framework.

- (4) α represents the attenuation coefficient of service profit of logistics orders in the transportation process. The value range of α is (0, 1). The reason for introducing the profit attenuation factor is that the long-term benefit has a certain uncertainty. Once the transportation time exceeds the preset time threshold, the profit $\sum_{j=1}^l p_i^j$ will decrease exponentially with the accumulation of the timeout time according to the attenuation coefficient α .

During the interaction between the environment and the agent, the $t-1$ moment environment will select the appropriate logistics orders and obtain their status s_{t-1} to transmit to the agent. Based on this information, the agent will select the legitimate scheduling action a_{t-1} from the feasible action according to its decision strategy and sample it. After the scheduling scheme is formed, the agent assigns the scheme to the selected orders in the s_{t-1} state for scheduling, and then the state transfers to s_t . At time t , the environment picks the right order again and repeats the process until all logistics orders are processed.

4. PTB model

To solve the flexible logistics service combination problem with spatial-temporal constraints, we construct the deep reinforcement learning framework as shown in Fig. 2. The framework consists of two parts, FLSCPS environment and the agent. The FLSCPS environment contains both static and dynamic information. Static information contains service constraints, resource status, and order attributes, while dynamic information contains order attributes, order processing progress, order selection times, and optimal service provider. Agents are further divided into variable length sequence module (GVLS) based on greedy strategy, policy network and value network. GVLS is responsible for generating real-time order status, and the policy network consists of the Transformer, stacked Bayes module, and Softmax classifier to output scheduling actions. The value network consists of stacked Bayesian convolution and stacked Bayesian linear layers that output the value of the current state. During the scheduling decision, FLSCPS environment transmits the initial state to GVLS, and after the GVLS processes the initial state, the real-time state is sent to the policy network. The policy network extracts the features layer by layer and selects the scheduling action. Then the scheduling action is fed back to FLSCPS environment, and the selected service is assigned to the order. When the scheduling is completed, the information such as actions and rewards in the scheduling process is collected, which is used to train the Agent by the reinforcement learning algorithm (PPO) and update the parameters of the policy network and the value network. Finally, the optimized scheduling strategy can be used to solve FLSCPS instances of different sizes to obtain satisfactory solutions.

The following is a detailed introduction to GVLS, policy network, value network and reinforcement learning training process.

4.1. GVLS module

Aiming at the process of generating state information, we adopt an optimized method. The process begins by randomly drawing a certain number of orders from the candidate order set. Then for each order of service, the greedy strategy is used to select the service provider that can provide the highest service efficiency for processing, and then connect the status information sequence of these orders (selection times and completion status). To fit the variable length sequence processing module, the resulting sequence needs to

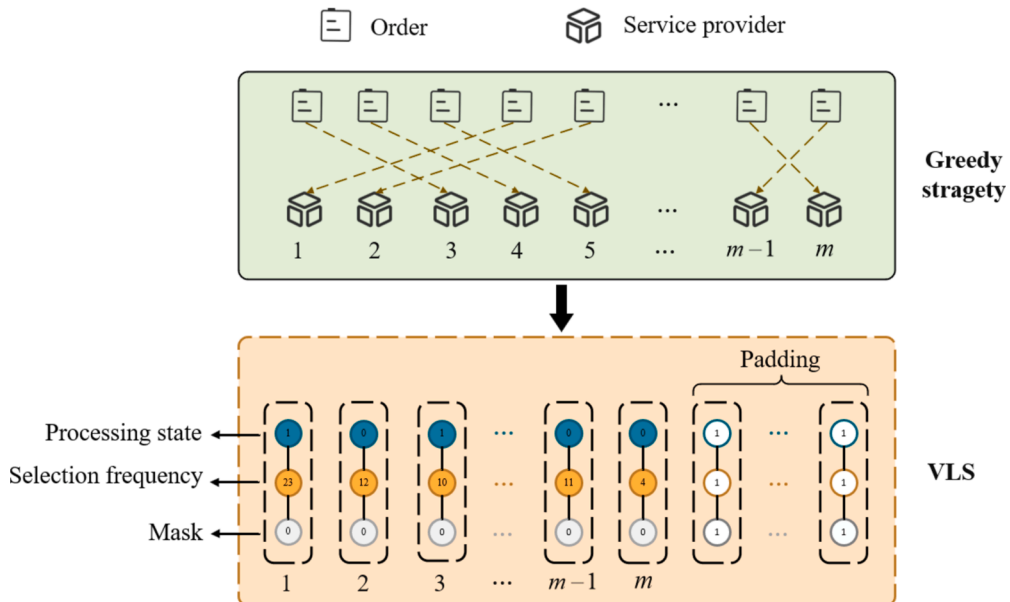


Fig. 3. Variable length sequence module based on greedy strategy.

be extended to the maximum sequence length. During this process, a sequence of masking vectors is introduced to label the filling position. Finally, the original state information sequence and the masking vector sequence are combined to form the base of the network input. Fig. 3 shows the flow of the GVLS module.

The greedy strategy of selecting the best service provider for the order works as follows. To efficiently allocate service providers for all orders, it is initially necessary to select a pending order for each service provider based on their capabilities. When selecting pending orders at time step t , priority should be given to those orders that have unprocessed services, and those orders should be assigned to the service provider with the shortest processing time. But if that service provider has already been assigned an order, the service provider with the second shortest processing time needs to be selected to process the unassigned order, and so on. In addition, once an order is assigned to a service provider, the order cannot be assigned to another service provider to avoid the waste of resources caused by repeated processing. Through the above optimization methods, m orders to be processed can be selected in the end. For m orders selected in time step t , obtain its selection times Nop_{o_i} and state Os_{o_i} to generate a $2 \times m$ matrix, and use GVLS to fill this matrix into a $3 \times m$ matrix $V_{t,m}$, where each row contains the selection times, state, and filling value of an order. The matrix $V_{t,m}$ is represented as follows.

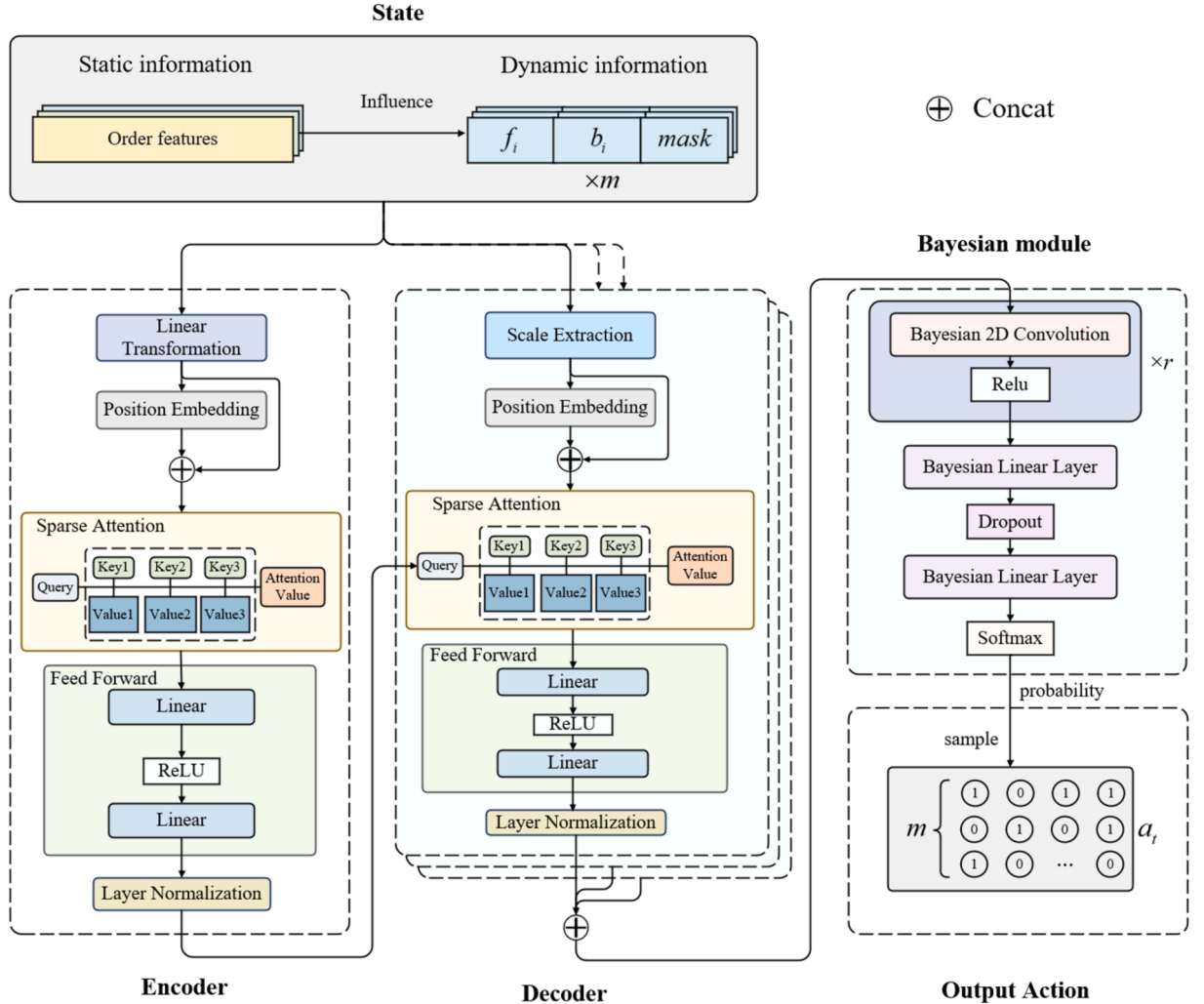


Fig. 4. Policy network structure.

$$V_{t,m} = \begin{bmatrix} Nop_{o_1} & Os_{o_1} & 0 \\ Nop_{o_2} & Os_{o_2} & 0 \\ \dots & \dots & \dots \\ Nop_{o_m} & Os_{o_m} & 0 \\ Nop_{o_{m+1}} & Os_{o_{m+1}} & 1 \\ \dots & \dots & \dots \\ Nop_{o_{mc}} & Os_{o_{mc}} & 1 \end{bmatrix} \quad (15)$$

After the matrix vector $V_{t,m}$ generated by the GVLS module is passed to the agent, the agent will return the decision action on whether to process the service of the m orders. For example, for the j -th service demand d_i^j of order o_i , the optimal service provider selected previously will decide whether to process the demand d_i^j according to the decision action and various constraints of the environment. When the demand for m orders has been processed, the transmission is transferred to the new state s_{t+1} . The following content mainly explains the principle of variable length sequence module.

Initially the maximum number of service providers in the model is set to mc , which can be dynamically adjusted for different situations. In the process of environment interaction, the corresponding m orders are first screened according to the number of service providers in the data set. Subsequently, the sequence $S = \{s_1, s_2, \dots, s_i, \dots, s_m\}$ of status information corresponding to these orders is then collected, which includes the processing status $F = \{f_1, f_2, \dots, f_i, \dots, f_m\}$ and the number of picks $B = \{b_1, b_2, \dots, b_i, \dots, b_m\}$. Based on the maximum number of service providers considered by the model, the length of the state information sequence that needs to be filled with 0 is calculated to be $(mc - m)$. For the shorter sequence, the 0-filling method is used to extend them to the length of the longest sequence, and then generate the adjusted state information sequence $S' = \{s_1, s_2, \dots, s_i, \dots, s_m, 0_{m+1}, \dots, 0_{mc}\}$. Meanwhile a masking vector sequence $Mask = \{0_1, 0_2, \dots, 0_i, \dots, 0_m, 1_{m+1}, \dots, 1_{mc}\}$ is introduced to identify the fill position, where 0 represents the original data and 1 represents the fill part.

In the complex FLSCPS environment, the variable length sequence module continuously responds to the environment changes and updates its data, while the static information also continuously influences the value of the variable length sequence module, making it change dynamically. Finally, the output of the variable-length sequence module is passed to the policy network to guide the final scheduling action.

4.2. Policy Net

The architecture of the policy network combines the Transformer and the Bayesian network, including Encoder-Decoder module, BayesCNN layer, BayesLinear layer and softmax activation function. Encoder-Decoder modules are used to encode and decode order data to extract and integrate global information and to capture features and long-distance connections of the input data. The BayesCNN layer is used to capture the spatial properties of the output features of the Encoder-Decoder module to further improve the accuracy of prediction. The BayesLinear layer is used to transform the output features into the dimensions required for the appropriate policy actions, and consider the uncertainty of the parameters, using Bayesian inference method for training and prediction. This step helps match the model output to the target action. Finally, the softmax activation function is used to convert the linear output into a probability distribution that can be used to decide the choice of action. Doing so allows the model to output a probability distribution for each action, thus selecting the optimal policy action more accurately.

The following is a detailed introduction to the data flow of Encoder module, Decoder module and Bayesian module in the policy network.

Encoder: The Encoder in the Fig. 4 has 3 layers ($u=3$). Suppose the current time step is t and the input is state information x (which is a $2 \times m$ matrix), which can be represented as a vector. then the data dimension of x is increased by linear transformation, $x \in \mathbb{R}^{d_{model}}$, where $d_{model} = 32$ represents the model dimension. x is then concatenated with the location code in order to complement the input data with the location information and increase the model's ability to represent the input data. The final output is x' , and the formula is as follows.

$$x' = Linear(x, W_0) + p_t \quad (16)$$

where W_0 represents learnable weights of the linear layer, p_t represents position coding, and $Linear$ represents linear transformations. For each input vector x'_i , first define a related index set Idx_i , which contains the indexes of all input vectors related to x'_i . For each index j in the set Idx_i , the attention score between x'_i and x'_j is calculated. If j is not in Idx_i , the score defaults to 0. The formula is as follows.

$$Attention(x'_i, x'_j) = \begin{cases} score(x'_i, x'_j), & \text{if } j \in Idx_i \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

where $score(x'_i, x'_j)$ stands for similarity measure. Next, the attention scores within the set Idx_i are normalized using the softmax function to get the attention weight. The formula is as follows.

$$Attention_Weight(x'_i, x'_j) = \frac{e^{Attention(x'_i, x'_j)}}{\sum_{k \in S_i} Attention(x'_i, x'_k)} \quad (18)$$

Finally, the calculated attention weights are used to weight the sum of the input vectors in set Idx_i to get the output vector SA_i . The formula is as follows.

$$SA_i = \sum_{j \in S_i} Attention_Weight(x'_i, x'_j) \times x'_j \quad (19)$$

Concatenate all the outputs SA_i of sparse attention to get a vector $A \in \mathbb{R}^{d_{model}}$, then add that vector to the original $Q(x)$, and get.

$$Q(x)' = Q(x) + A \quad (20)$$

$Q(x)'$ is normalized and obtained.

$$z_1 = LayerNorm(Q(x)') = \frac{Q(x)' - \mu}{\sqrt{\sigma^2 + \varepsilon}} \odot \gamma + \beta \quad (21)$$

where μ and σ are their mean and standard deviation, respectively, and ε is a small positive number used to ensure that the denominator is not zero. γ and β are the number of learnable scaling parameters and bias parameters, both vectors of the same dimension as the input $Q(x)'$. z_1 is then passed as input to a feedforward neural network for processing.

$$z_2 = ReLU(z_1 W_1 + b_1) W_2 + b_2 \quad (22)$$

where $W_1 \in \mathbb{R}^{d_{model} \times d_{ff}}$, $b_1 \in \mathbb{R}^{d_{ff}}$, $W_2 \in \mathbb{R}^{d_{ff} \times d_{model}}$ and $b_2 \in \mathbb{R}^{d_{model}}$ are the learnable weights and bias parameters, respectively. d_{ff} represents the dimension of the middle layer of forward propagation (also known as the hidden layer dimension of the fully connected layer); $ReLU$ stands for nonlinear activation function. And then add z_1 and z_2 to obtain.

$$z_3 = z_1 + z_2 \quad (23)$$

Layer Normalization of z_3 is performed to get the final output vector.

$$out = LayerNorm(z_3) = \frac{z_3 - \mu'}{\sqrt{\sigma'^2 + \varepsilon'}} \odot \gamma' + \beta' \quad (24)$$

The meaning of parameters is the same as that of equation (21). The output vector out returned by the above formula will be used as input K and V for sparse attention in the decoder module, and will be used as the calculation process for calculating the multi-head attention vector A' and the final output vector of the decoder output.

Decoder: There is one Decoder in the Fig. 4 ($v=1$). First the input data needs to be processed. The state information x is first transformed through the linear layer, then feature extraction is carried out through the convolution layer with different kernel sizes respectively, and then the input v_t of the three decoders is obtained after concatenated with the position coding. The kernel of the convolution layer is w_k , the kernel size is $k = 2$, and the convolution kernel output is $c_t^{(k)}$, then the input vector $v_t^{(1)}$ of the first Decoder can be expressed as.

$$v_t^{(1)} = Conv2D(Linear(x, W_3), w_2) + p_t \quad (25)$$

Among them, $Linear(x, W_3)$ represents the linear transformation of state information to obtain a vector whose dimension is the same as the size of the convolution kernel; $Conv2D$ represents the two-dimensional convolution operation; convolution kernel w_2 is used to convolve the linear transformed vector to obtain the result $c_t^{(1)}$ processed by the convolution layer. Finally, $c_t^{(1)}$ is concatenated with the position code p_t to obtain the input vector $v_t^{(1)}$ of the first Decoder.

Similarly, the input vector $v_t^{(2)}$ of the second Decoder can be represented as.

$$v_t^{(2)} = Conv2D(Linear(x, W_4), w_3) + p_t \quad (26)$$

where $Linear(x, W_4)$ is the same linear layer as in the input of the first Decoder, $v_t^{(2)}$ differs from $v_t^{(1)}$ in that the convolution kernel size is $k = 3$.

The input vector $v_t^{(3)}$ of the third Decoder can also be represented as.

$$v_t^{(3)} = Conv2D(Linear(x, W_5), w_4) + p_t \quad (27)$$

where the convolution kernel size is $k = 4$, and the rest are consistent with the meaning of equations (25) and (26). The v_t obtained above is the V vector, and the output of the Encoder is the Q and K vector of each Decoder module, respectively. After Q , K and V are passed into the Decoder, they first enter the sparse attention module. In the SPA module, the similarity or correlation score between the query vector and the key vector is calculated. Then the sparsity technique (mask mechanism) is applied to reduce the number of non-zero elements in the attention score, thus achieving sparsity. The sparse attention scores are then normalized to ensure that the sum of

the scores is 1. Finally, the normalized attention score is multiplied with the value vector to obtain the weighted value vector, and the final output A' of the multi-head attention module is obtained, as follows.

$$A' = \text{SparseAttention}(Q, K, V) \quad (28)$$

where *SparseAttention* represents the sparse attention layer, which is the same as formula (17)–(19).

Then the attention output result A' will directly enter the feedforward network layer, and the feedforward network layer will further process the attention result to obtain the output z_4 , as shown in formula (29).

$$z_4 = \text{FeedForward}(A') \quad (29)$$

where *FeedForward* indicates the feedforward network layer. Then the output z_4 of the feedforward network layer is residually connected with the attention result A' . Finally, the result obtained after the residual connection is normalized to obtain a Decoder final result y_t , which is detailed as follows.

$$y_t = \text{LayerNorm}(z_4 + A') \quad (30)$$

where *LayerNorm* represents the layer normalization operation, and y_t corresponds to the output of the three Decoder modules as $y_t^{(1)}$, $y_t^{(2)}$ and $y_t^{(3)}$ respectively.

Then the output y of the Transformer module can be obtained after connecting the outputs of the three decoders. The formula is as follows.

$$y = y_t^{(1)} + y_t^{(2)} + y_t^{(3)} \quad (31)$$

In summary, in the design of Decoder module, the sparse attention mechanism can effectively reduce the influence of noise, make the attention focus on meaningful elements in the input sequence, and improve the performance of the model, while the feedforward neural network can enhance the representation ability of the model through nonlinear transformation. In this way, the Decoder can combine the output of the Encoder with the input of the current time step to get a more accurate output result, thus achieving the sequence-to-sequence conversion task.

Bayes Module: Compared to the standard transformer, the output part of the PTB model has been changed. The standard the Transformer first performs a linear transformation after the Decoder output, and then uses the softmax activation function to get the probability distribution. Instead, we use 3 layers of Bayesian convolution ($r = 3$ in the Fig. 4), 2 layers of Bayesian full connection and 1-layer softmax to replace the final output module of the Transformer. The calculation process of this structure is as follows.

First, the final output y of the Decoder module is input into three different Bayesian convolution layers to extract the feature representation of the input information. For each Bayesian convolution layer, the input is the output result of the previous layer, and the final output is the result of the convolution operation, as follows.

$$y'_1 = \text{BayesConv}(y, W_6) \quad (32)$$

$$y'_2 = \text{BayesConv}(y'_1, W_7) \quad (33)$$

$$y'_3 = \text{BayesConv}(y'_2, W_8) \quad (34)$$

where *BayesConv* represents the Bayesian convolution layer, y'_1, y'_2, y'_3 and W_6, W_7, W_8 are the output and weight parameters of the first, second and third layers of Bayesian convolution, respectively.

Then the feature information of y'_3 is further extracted through two layers of Bayesian linear layers and the input is transformed into a 1×2 (action dim) vector, and the output result can be expressed as.

$$BF_1 = \text{BayesLinear}(y'_3, W_9) \quad (35)$$

$$BF_2 = \text{BayesLinear}(BF_1, W_{10}) \quad (36)$$

where *b* represents the Bayesian linear layer, BF_1, BF_2, W_9 and W_{10} are the output and weight parameters of the two layers of Bayesian linear layer respectively.

Finally, the output BF_2 of the linear layer is input into the softmax layer to output the probability distribution of the action, which can be expressed as.

$$Aprob = \text{softmax}(BF_2) \quad (37)$$

where *Aprob* represents the action probability distribution and *softmax* is the activation function.

The main reason the PTB model use the Bayesian networks instead of the Transformer's final output layer is to improve the predictive accuracy and generalization ability of the model. The Bayesian convolution and fully connected layers can extract multi-layer feature representations of input information to better capture order information, while the softmax layer can predict the probability distribution of order actions based on this feature information, improving the predictive power of the model.

4.3. Value Net

The value network adopts the structure of a four-layer Bayesian convolution and a two-layer Bayesian linear layer. The input state information is first extracted and integrated through four Bayesian convolution layers, which all use Bayesian methods for weight learning and uncertainty modeling to enhance the robustness of the model. Further nonlinear mapping and abstract representation learning of the features output from the convolutional layer are carried out through a Bayesian linear layer, which also uses Bayesian methods to learn the weights and modeling uncertainties, so as to better utilize the observed data and optimize the model parameters. Finally, the state information is mapped to an independent value through a Bayesian linear layer, which is used as a value function to estimate the expected return or value of a given state. Mapping the state information to a single value helps to effectively estimate and compare the value of the state, so as to make decisions and optimize.

The design of this multi-layer Bayesian structure can effectively learn and represent the value information of the state, and take into account the uncertainty and the interpretability of the model. Therefore, this structure helps to improve the estimation performance and generalization ability of value functions. Meanwhile because the Bayes module structure of value network and policy network are similar, their data flow process is similar.

4.4. DRL training process

In each round of model learning training, the environment is first initialized, the transition dictionary (which records the current state, next state, action and reward in the process of state transfer) and the initial state is extracted from the environment. Then GVLS processes the initial state and gets the real-time state s_t . After the state s_t is passed into the policy network, the policy network outputs the action probability. Then the Agent samples the output probability of the policy network m times to obtain action a_t corresponding to each service provider. Then the action is transferred to the environment, and after the environment executes the scheduling action, it gets the instant reward r_t , while the scheduling environment is transferred to the state s_{t+1} , and finally the interaction experience (s_t, a_t, r_t, s_{t+1}) between the Agent and the environment is stored in the transition dictionary. After a round of training, the interactive experience in the transition dictionary is extracted and used to update the policy network and value network parameters. The network update process is as follows.

First, the temporal difference object is calculated according to the interactive experience.

$$target_t = r_t + \gamma V_{\omega}(s_{t+1}) \quad (38)$$

Then the temporal difference error is calculated.

$$\delta_t = target_t - V_{\omega}(s_t) \quad (39)$$

Then the advantage function is calculated.

$$A_t = \delta_t + \gamma \lambda A_{t+1} \quad (40)$$

The value network is trained using the method of gradient descent and the loss function $L(\omega)$ is defined as follows.

$$L(\omega) = E \left[(r + \gamma V_{\omega}(s_{t+1}) - V_{\omega}(s_t))^2 \right] \quad (41)$$

PPO algorithm uses a form of PPO-clip when updating the policy network, which imposes restrictions in the objective function to ensure that the new parameters do not differ too much from the old ones. The policy network loss function $L(\theta)$ is defined as follows.

$$L(\theta) = E \left[- \min \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_k}(a|s)} A^{\pi_{\theta_k}}(s, a), \text{clip} \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_k}(a|s)}, 1 - \varepsilon, 1 + \varepsilon \right) A^{\pi_{\theta_k}}(s, a) \right) \right] \quad (42)$$

Meanwhile when the network parameters are updated, a sequence of data is used to train the number of rounds, and the network is updated epochs times after each round of training. By constantly updating the parameters of the policy network and the value network, the strategy is iteratively optimized to obtain the best logistics service combination scheme. The pseudocode for the training is as follows.

Algorithm 1. Learning algorithm of logistics service combination model based on PPO.

Algorithm 1.	Learning algorithm of logistics service combination model based on PPO
Input.	Initialize Environment Initialize transition dictionary D Initialize policy network $\pi_{\theta}(a s)$ Initialize value network $v_{\omega}(s)$
1.	For episodes = 1, 2, ..., MAX_EPISODE do
2.	For t = 1, 2, ..., nT do
3.	Get state s_t from the environment.
4.	The state s_t is passed into the Agent to return to the scheduling action a_t .
5.	Pass the scheduling action a_t into the environment to obtain the next state s_{t+1} and return the reward r_t .
	(continued on next page)

(continued)

Algorithm 1.	Learning algorithm of logistics service combination model based on PPO
6.	Store the single-step interaction sequence (s_t, a_t, r_t, s_{t+1}) into D .
7.	End For
8.	compute TD Target by formula (38).
9.	compute TD Error by formula (39).
10.	compute advantage function by formula (40).
11.	compute value network loss by formula (41).
12.	compute policy network loss by formula (42).
13.	update $\pi_\theta(a s)$ and $v_\omega(s)$ using the Adam optimizer with learning rates α_θ is and α_ω respectively.
14.	End For
Output.	Updated policy network $\pi_\theta(a s)$, updated value network $v_\omega(s)$.

5. Experiments

In this section, we will evaluate the performance of the PTB model through a series of experiments, which mainly focus on the following eight aspects: data sets, evaluation indicators, comparative experiments, generalization study, ablation study, robustness study, parameter study, and large-scale data validation. Through these comprehensive evaluations, the aim is to gain a comprehensive understanding of the performance and applicability of the PTB models under different conditions.

5.1. Data sets

The experimental data consists of three files. (1) Order data, as shown in Table 2. Order data includes order start time, order volume, order service set, response time, latest completion time and other attributes. The generation time of the order starts from $t = 20$, and the generation time interval of adjacent orders is guaranteed to be in the range of $\{0, 1, 2, 3, 4\}$, the volume range of the order is $\{20, 30, 40\}$, and the response time range of the order is $\{20, 30, 40\}$. The smaller the response time of the order, the more urgent the order is, and the earlier the latest completion time of the order will be. (2) The service provider processes the service time data, as shown in Table 3. The data table includes the time for each service provider to process all services, the time range for the service provider to process the receiving service is $\{1, 2, 3, 4\}$, the time range for the trunk service is $\{20, 21, 22, 23, 24, 25, 26\}$, and the time range for the distribution service is $\{3, 4, 5, 6, 7\}$. If a service cannot be processed, the time is expressed as $INVALID = 999999; 3$) The service details data is shown in Table 4. The service detail data includes the vehicle capacity, service type, service profit and other attributes of the service. The vehicle volume of the collection service is 100, the profit range is $50 \sim 100$, the vehicle volume of the trunk service is 300, the profit range is $50 \sim 2000$, and the vehicle volume of the distribution service is 50, the profit range is $100 \sim 300$. In the training stage, the experimental data are randomly generated, and the generated order data conforms to the data specification of the scheduling task. Meanwhile we set the attenuation coefficient of order profit $\alpha = 0.98$ during transportation.

It should be noted that when the number of orders is the same, the change in the number of carriers will affect the overall profit of a batch of orders. In FLSCPS environment, the profit of orders is obtained by random generation, and the range of profit generation is randomly selected within a certain range. If the number of service providers is different, meaning that there are more or fewer providers participating in the service order, the profit range of each carrier will be different, resulting in a different number of providers participating in the service order. The total profit of the final order will be different.

5.2. Evaluation indicators

In order to comprehensively and carefully evaluate the performance of the model on specific tasks and ensure that the performance characteristics of the model are captured from different perspectives to more accurately guide the optimization and improvement of the model, we propose five new metrics, which are described in the following sections.

5.2.1. Indicator STR

The first indicator is the timeout rate for the start processing time of orders (STR), which measures the degree of delay in order response and reflects the efficiency and reliability of logistics services in order processing timeliness. The lower STR value indicates that the algorithm can effectively start the order processing process in the given time and meet the timely processing requirements of

Table 2
Order data.

Number	Start time	Volume	Service sets	Response time	Last time
1	20	20	53	48	188
2	20	20	51,43,32	24	144
...	...	...	...	...	...
n	518	30	19	24	638

Table 3

Time data for the service provider to process the order service.

Service number	Carrier 1	Carrier 2	...	Carrier m
1	4	999999	...	1
2	999999	999999	...	21
...	...	...	...	...
$m*3$	999999	3	...	999999

Table 4

Service data.

Service number	Vehicle capacity	Transport type	Transport section	Profit
1	100	collect	A->C	94
2	300	trunk	A->B	1059
...	...	...	...	...
$m*3$	50	delivery	B->C	127

logistics services. The formula for calculating the indicator STR is as follows.

$$STR = \frac{\sum_{i=1}^n Osst_{o_i}}{n} \quad (43)$$

In formula (43), n represents the number of orders in the instance, and $Osst_{o_i}$ represents whether the start processing time of order o_i has timed out. $Osst_{o_i}$ is defined as follows.

$$Osst_{o_i} = \begin{cases} 1, & \text{if } T_{o_i}^{sp} - T_{o_i}^{generate} - T_{o_i}^{res} > 0 \\ 0, & \text{otherwise} \end{cases} \quad (44)$$

In formula (44), $T_{o_i}^{sp}$, $T_{o_i}^{generate}$, and $T_{o_i}^{response}$ represent the initial processing time, generation time, and response time of the order o_i respectively. We then set up several other variables to calculate the confidence interval and effect size of the indicator STR. $STR_{Comparative}$ represents the order start processing timeout rate calculated after all orders are processed by the traditional method. NoT_{STR} represents the order start processing time timeout rate obtained when a particular instance of order data was processed at the i -th test of the model. $STR_{instance,i}$ represents the order start processing time timeout rate obtained when a particular instance of order data was processed at the i test of the model. $\overline{STR}$ represents the average start processing time timeout rate of orders obtained by the PTB model after multiple tests, which is calculated by the following formula:

$$\overline{STR} = \frac{1}{NoT_{STR}} \sum_{i=1}^{NoT_{STR}} STR_{instance,i} \quad (45)$$

Then we can calculate the confidence interval for the STR indicator. Usually this involves calculating the standard deviation σ_{STR} and then using the normal distribution to determine the upper and lower bounds of the confidence interval.

$$\sigma_{STR} = \sqrt{\frac{1}{NoT_{STR}} \sum_i^{NoT_{STR}} (STR_{instance,i} - \overline{STR})^2} \quad (46)$$

$$STR_{CL,lower} = \overline{STR} - z \cdot \frac{\sigma_{STR}}{\sqrt{NoT_{STR}}} \quad (47)$$

$$STR_{CL,upper} = \overline{STR} + z \cdot \frac{\sigma_{STR}}{\sqrt{NoT_{STR}}} \quad (48)$$

In formulas (47) and (48), z is the Z-score corresponding to the desired confidence level, $STR_{CL,lower}$ is the lower bound of the confidence interval for the indicator STR, and $STR_{CL,upper}$ is the upper bound of the confidence interval for the indicator STR. When the algorithm processes different order data with the same scale and data distribution, it will generate different logistics service combination schemes, which leads to certain deviation of STR indicator value. To ensure the reliability of the evaluation results, we set a 95 % confidence level, corresponding to a Z-score of 1.96.

Finally, the effect size can be calculated in the following way.

$$\overline{STR}_{ES} = \frac{\overline{STR} - STR_{Comparative}}{STR_{Comparative}} \times 100\% \quad (49)$$

$\overline{STR}_{ES}$ Compared the advantages of the PTB model over comparative methods in order start processing time timeout rate. When the effect size of the indicator STR is negative, this indicates that the PTB model does not perform well in reducing the timeout rate at the

start of order processing, and actually increases the latency of order processing compared to the comparison method. In contrast, a positive effect size means that the PTB model is superior to the comparison method in reducing the order timeout rate, successfully reducing the processing delay, and thus improving the efficiency of order processing.

5.2.2. Indicator ETR

The second indicator is the timeout rate for the end processing time of orders (ETR), which is a key indicator to measure the timeliness of logistics services in the order completion stage, and it reflects the efficiency and reliability of logistics services in the entire process from the beginning of order processing to the end. A low ETR value indicates that the logistics service is able to complete the order processing within the predetermined time, thus meeting the customer's expectation of logistics timeliness. The calculation formula of indicator ETR is as follows.

$$ETR = \frac{\sum_{i=1}^n Oest_{o_i}}{n} \quad (50)$$

In formula (50), n indicates the number of orders in the instance, and $Oest_{o_i}$ indicates whether the end processing time of order o_i has timed out. $Oest_{o_i}$ is defined as follows.

$$Oest_{o_i} = \begin{cases} 1, & \text{if } T_{o_i}^{ep} - T_{o_i}^{generate} - T_{o_i}^{mep} > 0 \\ 0, & \text{otherwise} \end{cases} \quad (51)$$

In formula (51), $T_{o_i}^{ep}$, $T_{o_i}^{generate}$, and $T_{o_i}^{latest}$ represent the end processing time, generation time, and latest processing completion time of order o_i respectively. We then set up several other variables to calculate the confidence interval and effect size of the indicator ETR. $ETR_{Comparative}$ represents the timeout rate for the end processing time calculated after all orders are processed by traditional methods. NoT_{ETR} represents the number of tests performed on the dataset after the model is trained. $ETR_{instance,i}$ represents the timeout rate for the end processing time obtained when a particular instance of order data was processed at the i -th test of the model. $\overline{ETR}$ represents the average end processing timeout rate of orders obtained by the PTB model after multiple tests, which is calculated by the following formula.

$$\overline{ETR} = \frac{1}{NoT_{ETR}} \sum_{i=1}^{NoT_{ETR}} ETR_{instance,i} \quad (52)$$

The confidence interval calculation method of indicator ETR is the same as that of indicator STR. Both of them need to calculate the calculated standard deviation σ_{ETR} first, and then use normal distribution to determine the upper and lower bounds of the confidence interval.

$$\sigma_{ETR} = \sqrt{\frac{1}{NoT_{ETR}} \sum_{i=1}^{NoT_{ETR}} (ETR_{instance,i} - \overline{ETR})^2} \quad (53)$$

$$ETR_{CL,lower} = \overline{ETR} - z \cdot \frac{\sigma_{ETR}}{\sqrt{NoT_{ETR}}} \quad (54)$$

$$ETR_{CL,upper} = \overline{ETR} + z \cdot \frac{\sigma_{ETR}}{\sqrt{NoT_{ETR}}} \quad (55)$$

In formula (54) and (55), z is the Z-score corresponding to the desired confidence level, $ETR_{CL,lower}$ is the lower bound of the confidence interval for the indicator ETR, and $ETR_{CL,upper}$ is the upper bound of the confidence interval for the indicator ETR. When the algorithm processes different order data with the same scale and data distribution, it will generate different logistics service combination schemes, which leads to a certain deviation of ETR indicator value. To ensure the reliability of the evaluation results, we set a 95 % confidence level, corresponding to a Z-score of 1.96.

Finally, the effect size can be calculated in the following way.

$$\overline{ETR}_{ES} = \frac{\overline{ETR} - ETR_{Comparative}}{ETR_{Comparative}} \times 100\% \quad (56)$$

$\overline{ETR}_{ES}$ Compared the advantages of the PTB model over comparative methods in the timeout rate for the end processing time of orders. When $\overline{ETR}_{ES}$ is positive, it indicates that the PTB model has achieved significant improvement in order processing speed compared with the comparison method, effectively reducing the order delay time. Conversely, if $\overline{ETR}_{ES}$ is negative, it means that the PTB model actually increases the delay in end-of-order processing, resulting in an overall decrease in efficiency.

5.2.3. Indicator TP

The third indicator is total profit (TP), which reflects the profitability of logistics service providers under the cooperative mode and is a key indicator to measure their business efficiency and profit performance. The purpose of this indicator is to quantify and evaluate the economic benefits of logistics service providers under specific logistics order instances. We set other variables to calculate the

confidence interval and effect size of the indicator TP. $TP_{Comparative}$ represents the total profit from all orders processed by traditional methods. NoT_{TP} represents the number of tests performed on the data set after the model training is completed. $TP_{instance,i}$ represents the profit from processing a particular instance of order data the i -th time the algorithm is run. $\overline{TP}$ represents the average total profit obtained by the PTB model after multiple tests, which is calculated by the following formula.

$$\overline{TP} = \frac{1}{NoT_{TP}} \sum_{i=1}^{NoT_{TP}} TP_{instance,i} \quad (57)$$

The confidence interval calculation method of indicator TP is the same as that of indicator STR. Both of them need to calculate the calculated standard deviation σ_{TP} first, and then use normal distribution to determine the upper and lower bounds of the confidence interval.

$$\sigma_{TP} = \sqrt{\frac{1}{NoT_{TP}} \sum_{i=1}^{NoT_{TP}} (TP_{instance,i} - \overline{TP})^2} \quad (58)$$

$$TP_{CL,lower} = \overline{TP} - z \cdot \frac{\sigma_{TP}}{\sqrt{NoT_{TP}}} \quad (59)$$

$$TP_{CL,upper} = \overline{TP} + z \cdot \frac{\sigma_{TP}}{\sqrt{NoT_{TP}}} \quad (60)$$

In formula (59) and (60), z is the Z-score corresponding to the desired confidence level, $TP_{CL,upper}$ is the lower bound of the confidence interval for the indicator TP, and $TP_{CL,lower}$ is the upper bound of the confidence interval for the indicator TP. When the algorithm processes different order data with the same scale and data distribution, there is little difference in the overall profit of the generated logistics service portfolio schemes, so we set a confidence level of 99 %, corresponding to a Z-score of 2.576.

Finally, we can calculate the effect size of the PTB model relative to comparative methods.

$$TP_{ES} = \frac{\overline{TP} - \overline{TP}_{Traditional}}{\overline{TP}_{Traditional}} \times 100\% \quad (61)$$

In formula (58), TP_{ES} represents the effect size of the PTB model relative to the traditional method. Based on the value of the percentage of effect size, we can quantitatively evaluate the relative advantages and disadvantages of the PTB model compared with traditional methods in terms of economic benefits, and specifically determine the degree of its advantages or disadvantages. Positive TP_{ES} indicates that the PTB model can achieve higher overall profits than the comparison method, reflecting its superiority in economic benefits. A negative value of TP_{ES} means that the PTB model performs poorly in the overall profit, which is lower than the comparison method, showing its deficiency in economic benefits.

5.2.4. Indicator TTS

The fourth indicator is total time steps (TTS), which reflects the efficiency of logistics service processes, especially when dealing with time-sensitive logistics tasks, this indicator is particularly important. The purpose of putting forward the indicator TTS is to quantify the total time required to complete the entire logistics service process, and then we set other variables to calculate the confidence interval and effect size of the indicator TTS. $TTS_{Comparative}$ represents the total time step calculated after all orders are processed by the traditional method. NoT_{TTS} represents the number of tests performed on the data set after the model training is completed. $TTS_{instance,i}$ represents the end-of-order processing timeout rate obtained when a particular instance of order data was processed at the i -th test of the model. $\overline{TTS}$ represents the average total time step obtained by the PTB model after multiple tests, which is calculated by the following formula.

$$\overline{TTS} = \frac{1}{NoT_{TTS}} \sum_{i=1}^{NoT_{TTS}} TTS_{instance,i} \quad (62)$$

The confidence interval calculation method of indicator TTS is the same as that of indicator STR. Both of them need to calculate the calculated standard deviation σ_{TTS} first, and then use normal distribution to determine the upper and lower bounds of the confidence interval.

$$\sigma_{TTS} = \sqrt{\frac{1}{NoT} \sum_{i=1}^{NoT_{TTS}} (TTS_{instance,i} - \overline{TTS})^2} \quad (63)$$

$$TTS_{CL,lower} = \overline{TTS} - z \cdot \frac{\sigma_{TTS}}{\sqrt{NoT_{TTS}}} \quad (64)$$

$$TTS_{CL,upper} = \overline{TTS} + z \cdot \frac{\sigma_{TTS}}{\sqrt{NoT_{TTS}}} \quad (65)$$

In formula (64) and (65), z is the Z-score corresponding to the desired confidence level, $TTS_{CL,lower}$ is the lower bound of the confidence interval for the indicator TTS, and $TTS_{CL,upper}$ is the upper bound of the confidence interval for the indicator TTS. Since the time steps of the logistics service combination schemes generated by the algorithm are not very different when processing different order data with the same scale and data distribution, we set a confidence level of 99 %, corresponding to a Z-score of 2.576.

Finally, the effect size can be calculated in the following way.

$$\overline{TTS}_{ES} = \frac{\overline{TTS} - TTS_{Comparative}}{TTS_{Comparative}} \times 100\% \quad (66)$$

$\overline{TTS}_{ES}$ compares the magnitude of the advantage of the PTB model over traditional methods in terms of total time steps, which helps us understand the strength of the differences or associations between algorithms. When the value of $\overline{TTS}_{ES}$ is negative, it means that the PTB model can achieve shorter logistics order transportation time than the comparison method, and improve the efficiency of logistics transportation. When the value of $\overline{TTS}_{ES}$ is positive, it indicates that the transportation time of the PTB model is longer than that of the comparison method, but the logistics transportation efficiency decreases.

5.2.5. Indicator CST

The fifth metric is the algorithm computational solution time (CST), which represents the actual time it takes the algorithm to solve a given instance. In logistics service portfolio problems, algorithms need to be able to quickly respond to problems of different sizes and find solutions, so low CST becomes a key factor in achieving this goal. We then set up other variables to calculate the confidence interval and effect size of the indicator CST. $CST_{Comparative}$ represents the solution time calculated after all orders are processed by the comparative method. NoT_{CST} represents the number of tests performed on the data set after the model training is completed. $CST_{instance,i}$ represents the solution time to process a particular instance of order data was processed at the i -th test of the model. $\overline{CST}$ represents the average solution time of the PTB model after multiple tests, which is calculated by the following formula.

$$\overline{CST} = \frac{1}{NoT_{CST}} \sum_{i=1}^{NoT_{CST}} CST_{instance,i} \quad (67)$$

The confidence interval calculation method of indicator CST is the same as that of indicator STR. Both of them need to calculate standard deviation σ_{CST} first, and then use normal distribution to determine the upper and lower bounds of the confidence interval.

$$\sigma_{CST} = \sqrt{\frac{1}{NoT_{CST}} \sum_i (CST_{instance,i} - \overline{CST})^2} \quad (68)$$

$$CST_{CL,lower} = \overline{CST} - z \cdot \frac{\sigma_{CST}}{\sqrt{NoT}} \quad (69)$$

$$CST_{CL,upper} = \overline{CST} + z \cdot \frac{\sigma_{CST}}{\sqrt{NoT}} \quad (70)$$

In formula (69) and (70), z is the Z-score corresponding to the desired confidence level, $CST_{CL,lower}$ is the lower bound of the confidence interval for the indicator CST, and $CST_{CL,upper}$ is the upper bound of the confidence interval for the indicator CST. When the algorithm processes different order data with the same scale and data distribution, it will generate different logistics service combination schemes, which leads to a certain deviation of CST indicator value. To ensure the reliability of the evaluation results, we set a 95 % confidence level, corresponding to a Z-score of 1.96.

Finally, the effect size can be calculated in the following way.

$$\overline{CST}_{ES} = \frac{\overline{CST} - CST_{Comparative}}{CST_{Comparative}} \times 100\% \quad (71)$$

$\overline{CST}_{ES}$ compared the advantages of the PTB model in solving time compared with traditional methods. When $\overline{CST}_{ES}$ is negative, it indicates that the PTB model exceeds the comparison method in computation and processing speed, and can provide solutions at a faster speed. On the contrary, when $\overline{CST}_{ES}$ is positive, it means that the PTB model is inferior to the comparison method in terms of computation and processing speed, and takes longer time to complete the solution process.

Comparison experiments evaluate the differences between algorithms using these five metrics. Meanwhile we set four experimental scenes. Scene1 means that the number of services provided in the dataset is 20, Scene2 means that the number of services provided in the dataset is 25, Scene3 means that the number of services provided in the dataset is 30, and Scene4 means that the number of services provided in the dataset is 35. In these scenes, the number of logistics orders is not fixed.

5.3. Comparative experiment

Since FLSCPS has not yet been studied, Comparison experiments used five baseline methods to compare the PTB model. These include Earliest Deadline First (EDF), First Come First Served (FCFS), and Non-Complete Greedy. (NCG), PG (Policy Gradient) and DQN

(Deep Q-Network). The models were executed on a CPU server with a 2.50 GHz Intel i5-7300HQ CPU. The procedure for solving each baseline method is as follows.

The EDF algorithm assigns a priority to each order that has not yet been processed, and the order closer to the deadline will be given a higher priority and will be prioritized. The order is then assigned to the available service provider based on the order's priority, and this is repeated until all orders are processed.

The FCFS algorithm will process the order according to the generation time, the order generated first will be processed first, and the order generated later will be queued. Orders are processed in accordance with these rules at each step until all orders have been processed.

The NCG algorithm selects the uncompleted order that is closest to the deadline at each step, and selects a service provider that can process the order service with the shortest processing time. After selecting the service provider, the selected order and the corresponding service provider information are transmitted to the environment for processing. Repeat until all orders are processed.

PG algorithm uses the policy gradient method to optimize the policy parameters of the agent, obtains training data through the interaction between the agent and the environment, and obtains the expected return and the action reward, so as to achieve the goal of maximizing the expected return. In the gradient calculation formula, the difference between the return and the state value function is used to update the agent's policy parameters to improve the final performance.

DQN algorithm (Aghili et al., 2021) is an algorithm that combines deep learning and reinforcement learning, and deals with high-dimensional state space and complex action space by using neural networks to approximate Q-value functions. DQN uses experience replay and target network technology to improve the stability and efficiency of training. In the training process, by interacting with the environment, the weights of the Q network are constantly updated to learn the optimal policy.

Because FLSCPS processes different types of order services at each time node, and it also needs to be solved quickly. However, the solution time of planners such as OR-Tools far exceeds the need of real-time calculation in practical applications, so the PTB model is no longer compared with the solution results of planners in comparison experiments. Meanwhile the solution of the planner will not be discussed in the following experiments.

In the experiment, the PTB model and EDF, FCFS, NCG, PG and DQN algorithms are compared in four logistics order scenes, then randomly generated 100 instances for each set of data, and took the average of multiple calculation results as the final test result, as shown in Table 5.

By observing the data in Table 5, we conduct in-depth analysis on the performance of the PTB model and other algorithms in different scenes. First, from the perspective of the key indicator TP, the PTB model performs well in all scenes, especially in scene 3 and scene 4, where the total profit of the PTB model is significantly higher than that of other algorithms. For example, in scene 4, when the number of orders reaches 1000 pieces, the PTB model shows an effect size of 641 % in $\overline{TP}$ indicator compared with the EDF algorithm, which not only highlights the significant advantages of the PTB model in terms of economic benefits, but also reflects its ability to create a large difference in economic benefits. This result further proves that the PTB model has significant advantages in dealing with large-scale data processing and complex problem solving, and can provide more efficient solutions. Secondly, from the perspective of indicator $\overline{TTS}$, the PTB model requires less time step in most scenes, which shows its ability to quickly find the approximate optimal solution. In all scenes, the PTB model not only achieved the highest total profit, but also required the smallest time step, further demonstrating its significant efficiency advantage in solving complex optimization problems. This result highlights the excellent performance of the PTB model in the balance of solution quality and computational efficiency. Moreover, in terms of indicator $\overline{CST}$, the calculation time of the PTB model in all scenes is lower than that of other algorithms, showing its obvious advantages in computational efficiency, which can find the approximate optimal solution more quickly and improve the efficiency of problem solving. In scene 4, when the number of orders reached 1000, we obtained through statistical analysis that the 95 % confidence interval of the computing time of the PTB model was [8.56, 9.61], indicating that its advantage in computing efficiency was statistically significant. Meanwhile the data in the Table 5 also show that the NCG algorithm is basically equal to the PTB model in terms of total profit, which is attributed to its ability to approach the global optimal solution through local optimal selection. However, in terms of solving time, NCG algorithm is obviously inferior to the PTB model, because more time is needed for data processing and calculation when dealing with large-scale problems, resulting in longer solving time. Taking the above analysis results into consideration, we can conclude that the PTB model shows excellent performance in all scenes, no matter in terms of total profit, time step or calculation time, it is superior to other algorithms. This fully confirms the validity of our model design and provides a new and efficient method for solving large-scale optimization problems.

In order to further verify whether the PTB model can obtain better processing efficiency when processing different order sizes, we compared the two indicators of $\overline{STR}$ and $\overline{ETR}$ among the models in four scenes. The $\overline{STR}$ indicator assesses how responsive the model is when processing orders, while the $\overline{ETR}$ indicator assesses how timely the model is when processing orders. In Fig. 5, the horizontal axis represents the order size and the vertical axis represents the value of $\overline{STR}$; In Fig. 6, the horizontal axis represents the order size and the vertical axis represents the $\overline{ETR}$ value. A smaller value for $\overline{STR}$ and $\overline{ETR}$ means that the model performs better in terms of responsiveness and timeliness of order processing.

Based on the curves and values in Fig. 5 and Fig. 6, it can be seen that the PTB model shows the best processing efficiency in various scenes, while maintaining the lowest $\overline{STR}$ and $\overline{ETR}$ indicator values, indicating greater responsiveness and timeliness. As can be seen from the curve trend of the NCG algorithm in the figure, the processing efficiency of the NCG algorithm is the closest to that of the PTB model among all comparison algorithms. It can also be observed from the data in Table 5 that the NCG algorithm is equal to the PTB model in terms of total profit, while total profit is inversely proportional to the transportation time, that is, the greater the total profit, the shorter the transportation time. Reduced haulage time mean orders are processed more efficiently, so the NCG algorithms are close

Table 5

Performance comparison of different models on all scenes (Bold is optimal, * is suboptimal).

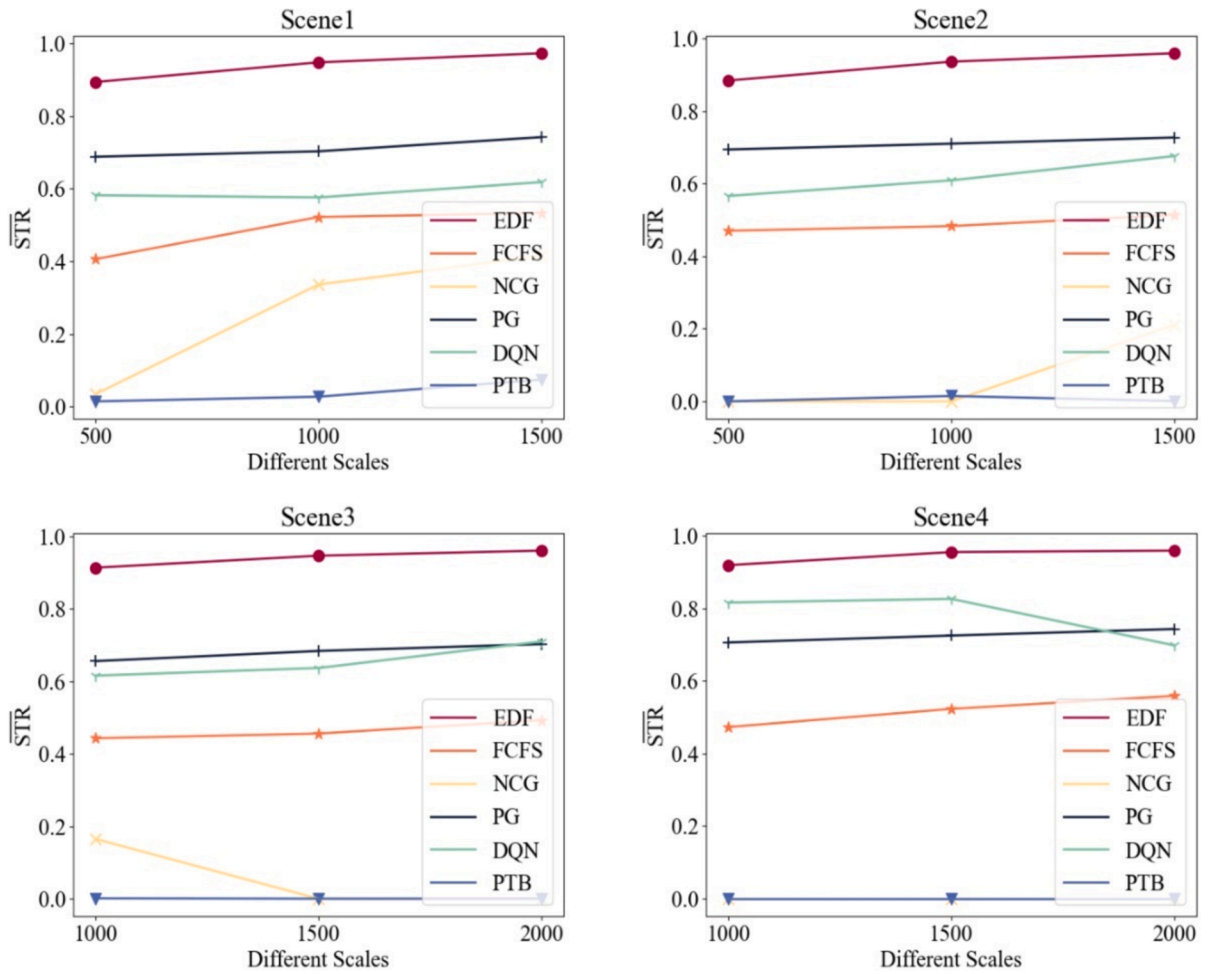
Scene	<i>n</i>	Indicator	EDF	FCFS	NCG	PG	DQN	PTB (ours)
Scene1	500	$\overline{TP}$	76922	292232	423678*	146392	204566	423678
		$\overline{TP}_{ES}$	450.79 %	44.98 %	0.00 %	189.41 %	107.11 %	–
		$\overline{TTS}$	1164	538	368*	4016	5201	344
		$\overline{TTS}_{ES}$	–70.45 %	–36.06 %	–6.52 %	–91.43 %	–93.39 %	–
		$\overline{CST}$	5.9*	6.2	18.10	11.91	43.64	5.83
		$\overline{CST}_{ES}$	–1.19 %	–5.97 %	–67.79 %	–51.05 %	–86.64 %	–
		$\overline{TP}$	86001	487898	869624*	251707	367120	870299
		$\overline{TP}_{ES}$	911.96 %	78.38 %	0.08 %	245.76 %	137.06 %	–
		$\overline{TTS}$	2363	996	656*	13708	13948	588
	1000	$\overline{TTS}_{ES}$	–75.12 %	–40.96 %	–10.37 %	–95.71 %	4423.08 %	–
		$\overline{CST}$	16.78	10.25*	67.36	53.16	136.62	9.91
		$\overline{CST}_{ES}$	–40.94 %	–3.32 %	–85.29 %	–81.36 %	–92.75 %	–
		$\overline{TP}$	76077	644747	1296777*	346954	499366	1298656
		$\overline{TP}_{ES}$	1607.03 %	101.42 %	0.14 %	274.30 %	160.06 %	–
		$\overline{TTS}$	3488	1402	968*	17849	17849	897
		$\overline{TTS}_{ES}$	–74.28 %	–36.02 %	–7.33 %	–94.97 %	–94.97 %	–
		$\overline{CST}$	35.07	13.25	145.60	88.58	199.98	16.53*
		$\overline{CST}_{ES}$	–52.87 %	24.75 %	–88.65 %	–81.34 %	–91.73 %	–
	1500	$\overline{TP}$	91721	313781	461355*	159280	234113	461355
		$\overline{TP}_{ES}$	403.00 %	47.03 %	0.00 %	189.65 %	97.07 %	–
		$\overline{TTS}$	1059	563	320*	7545	5895	320
		$\overline{TTS}_{ES}$	–69.78 %	–43.16 %	0.00 %	–95.76 %	–94.57 %	–
		$\overline{CST}$	49.83	16.75	9.30*	26.63	63.46	7.82
		$\overline{CST}_{ES}$	–84.31 %	–53.31 %	–15.91 %	–70.63 %	–87.68 %	–
		$\overline{TP}$	94997	509648	929783*	268688	359901	929783
		$\overline{TP}_{ES}$	878.75 %	82.44 %	0.00 %	246.05 %	158.34 %	–
		$\overline{TTS}$	2172	1040	561	14961	5916	611*
Scene2	500	$\overline{TTS}_{ES}$	–71.87 %	–41.25 %	8.91 %	95.91 %	–89.67 %	–
		$\overline{CST}$	35.26*	49.66	36.28	70.33	73.74	14.35
		$\overline{CST}_{ES}$	–59.30 %	–71.10 %	–60.45 %	–79.60 %	–80.54 %	–
		$\overline{TP}$	95935	660980	1379369*	338651	510099	1380910
		$\overline{TP}_{ES}$	1339.42 %	108.92 %	0.11 %	307.77 %	170.71 %	–
		$\overline{TTS}$	3128	1473	824	31090	35197	992*
		$\overline{TTS}_{ES}$	–68.29 %	–32.65 %	20.39 %	–96.81 %	–97.18 %	–
		$\overline{CST}$	101.04	59.57*	84.51	87.74	484.74	24.65
		$\overline{CST}_{ES}$	–75.60 %	–58.62 %	–70.83 %	–71.91 %	–94.91 %	–
	1000	$\overline{TP}$	123830	534538	889974*	293193	381621	889974
		$\overline{TP}_{ES}$	618.71 %	66.49 %	0.00 %	203.55 %	133.21 %	–
		$\overline{TTS}$	1860	947	443	13064	8028	514*
		$\overline{TTS}_{ES}$	–72.37 %	–45.72 %	16.03 %	–96.07 %	–93.60 %	–
		$\overline{CST}$	40.74	33.79	30.64*	71.65	115.86	8.034
		$\overline{CST}_{ES}$	–80.28 %	–76.22 %	–73.78 %	–88.79 %	–93.07 %	–
		$\overline{TP}$	131827	708985	1319792*	413577	534301	1319792
		$\overline{TP}_{ES}$	901.15 %	86.15 %	0.00 %	219.12 %	147.01 %	–
		$\overline{TTS}$	2723	1356	681	24976	10074	800*
	1500	$\overline{TTS}_{ES}$	–70.62 %	–41.00 %	17.47 %	–96.80 %	–92.06 %	–
		$\overline{CST}$	110.46	40.03	66.19*	177.19	164.78	19.39
		$\overline{CST}_{ES}$	–82.45 %	–51.56 %	–70.71 %	–89.06 %	–88.23 %	–
		$\overline{TP}$	124507	887759	1753012*	483985	603614	1753012
		$\overline{TP}_{ES}$	1307.96 %	97.46 %	0.00 %	262.20 %	190.42 %	–
		$\overline{TTS}$	3611	1738	872	36545	23553	1019*
		$\overline{TTS}_{ES}$	–71.78 %	–41.37 %	16.86 %	–97.21 %	–95.67 %	–
		$\overline{CST}$	190.75	77.07*	116.42	298.73	419.12	27.57
		$\overline{CST}_{ES}$	–85.55 %	–64.23 %	–76.32 %	–90.77 %	–93.42 %	–
Scene3	1000	$\overline{TP}$	121993	511179	904212*	271977	324539	904212
		$\overline{TP}_{ES}$	641.20 %	76.89 %	0.00 %	232.46 %	178.61 %	–
		$\overline{TTS}$	1689	876	417*	11934	898	395
		$\overline{TTS}_{ES}$	–76.61 %	–54.91 %	–5.28 %	–96.69 %	56.01 %	–
		$\overline{CST}$	47.68	8.29	23.68	75.11	110.91	9.17*
		$\overline{CST}_{ES}$	–74.79 %	44.99 %	–49.24 %	–84.00 %	–89.16 %	–
		$\overline{TP}$	122658	642601	1345388*	365578	444747	1345388
		$\overline{TP}_{ES}$	996.86 %	109.37 %	0.00 %	268.02 %	202.51 %	–
	1500							
Scene4	1000							
	1500							

(continued on next page)

Table 5 (continued)

Scene	<i>n</i>	Indicator	EDF	FCFS	NCG	PG	DQN	PTB (ours)
	2000	$\overline{TTS}$	2530	1329	610*	16596	14402	563
		$\overline{TTS}_{ES}$	-77.75 %	-57.64 %	-7.70 %	-96.61 %	-96.09 %	—
		$\overline{CST}$	86.54	40.82	53.33*	133.74	130.25	14.91
		$\overline{CST}_{ES}$	-82.77 %	-63.47 %	-72.04 %	-88.85 %	-88.55 %	—
		$\overline{TP}$	115518	783453	1785823*	437227	623614	1785823
		$\overline{TP}_{ES}$	1445.93 %	127.94 %	0.00 %	308.44 %	186.37 %	—
		$\overline{TTS}$	3322	1812	777*	33825	14402	707
		$\overline{TTS}_{ES}$	-78.72 %	-60.98 %	-9.01 %	-97.91	-95.09 %	—
		$\overline{CST}$	167.28	69.02*	94.20	319.48	403.12	19.21
		$\overline{CST}_{ES}$	-88.52 %	-72.17 %	-79.61 %	-93.99 %	-95.23 %	—
Total	Opt-Sub		0-2	2-4	5-23	0-0	0-0	29-7

* Notes: The statistical results of Opt and Sub are analyzed based on the data of indicators $\overline{TP}$, $\overline{TTS}$ and $\overline{CST}$, while the relevant results of indicators $\overline{TP}_{ES}$, $\overline{TTS}_{ES}$ and $\overline{CST}_{ES}$ are not included in the statistical scope of Opt and Sub.

Fig. 5. $\overline{STR}$ indicator algorithm comparison.

to the PTB model on $\overline{STR}$ and $\overline{ETR}$ indicators. Nevertheless, the PTB model still performs optimally on these two indicators. The results show that the PTB algorithm has good processing efficiency and excellent performance in handling various order sizes, and provides reliable support for the actual transportation process.

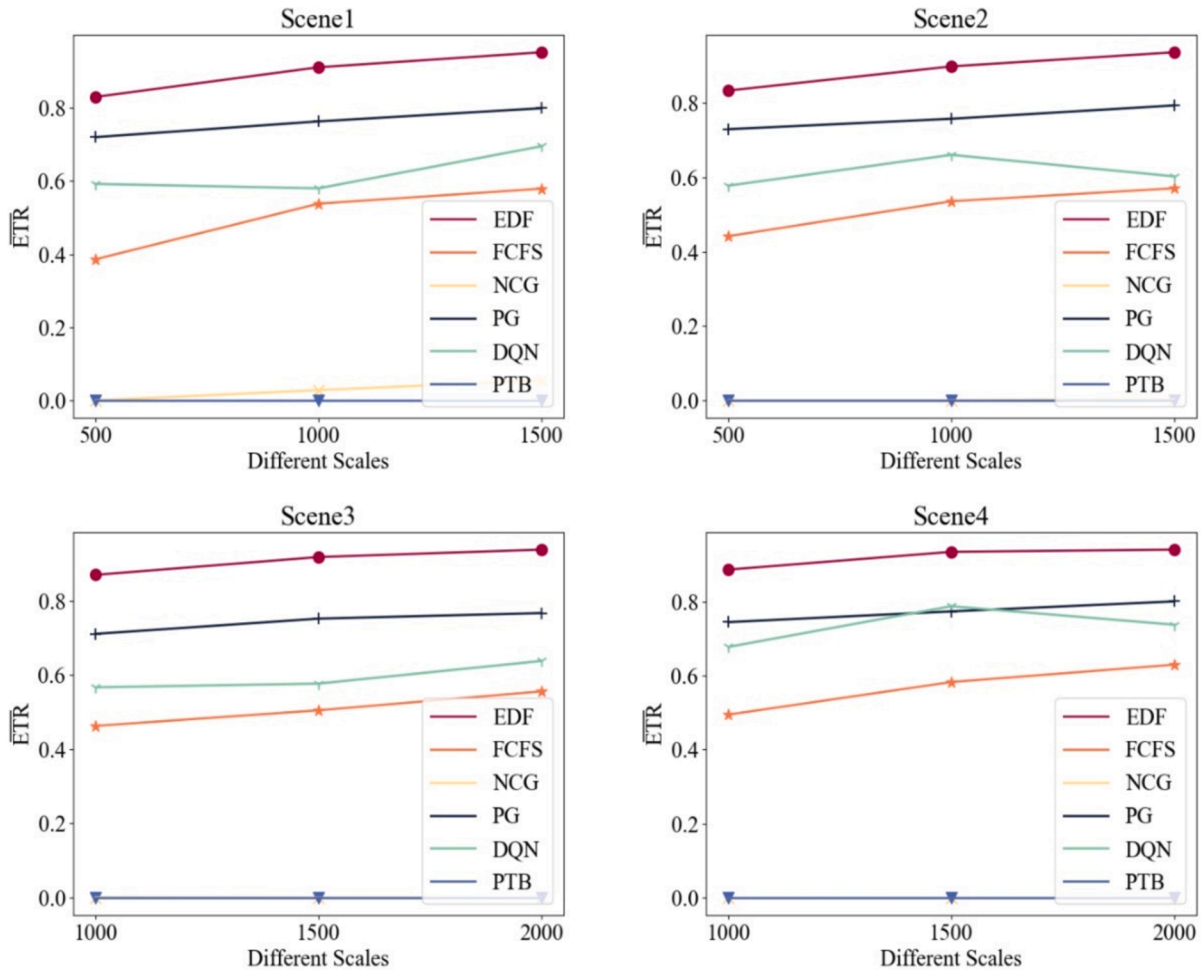


Fig. 6. ETR indicator algorithm comparison.

5.4. Generalization study

5.4.1. Study on model performance under different order sizes

In order to further verify the effectiveness and generalization ability of the PTB model to solve different order sizes, the problems of transporting 800, 1200, 1600, 2000, 2400 and 2800 logistics orders under four scenes are selected experimentally in this section. In Table 5, the PTB model achieves better overall results than the traditional heuristic algorithm and other comparison algorithms, and also performs well compared with NCG on large-scale problems, and the calculation time is satisfactory. Therefore, the NCG algorithm with the best performance in all aspects of the five comparison schemes is selected as the reference for verifying the generalization of the PTB model in this section on different order sizes in this experiment. Then TP and CST are used to evaluate the performance difference between the PTB model and NCG algorithm, and the comparison results are shown in charts.

Fig. 7 shows the overall profit comparison between the PTB model and NCG algorithm in four scenes processing different order sizes. The horizontal coordinate represents the instance size, and the vertical coordinate represents the profit. As can be seen from Fig. 7, in scene 1, the overall profit achieved by the PTB model is higher than that of the NCG algorithm when the order size gradually increases. However, NCG algorithms are greedy, so there are two main reasons for the poor results. First, the number of service providers is smaller in Scene 1 compared to other scenes, resulting in an increase in the number of orders each service provider needs to process and a more complex allocation process. Secondly, NCG algorithm always chooses the best option at each step of selection, without considering the global or long-term interests, and this local optimal solution will lead to poor performance of the algorithm in the global scope. In scene 2, Scene 3 and scene 4, the overall profit of the PTB model is basically consistent with that of NCG algorithm. From the specific overall profit value, the profit obtained by the PTB model in each scene is basically the highest profit that can be obtained by the order. This shows that with the increase in the number of service providers, the PTB model has reached the highest profit under different order sizes, which also reflects that the PTB model has excellent performance in handling logistics problems of various sizes and complexities, and can provide a reliable reference for actual transportation problems.

Fig. 8 compares the time required for the PTB method and the NCG algorithm to achieve the overall profit in each scene. The

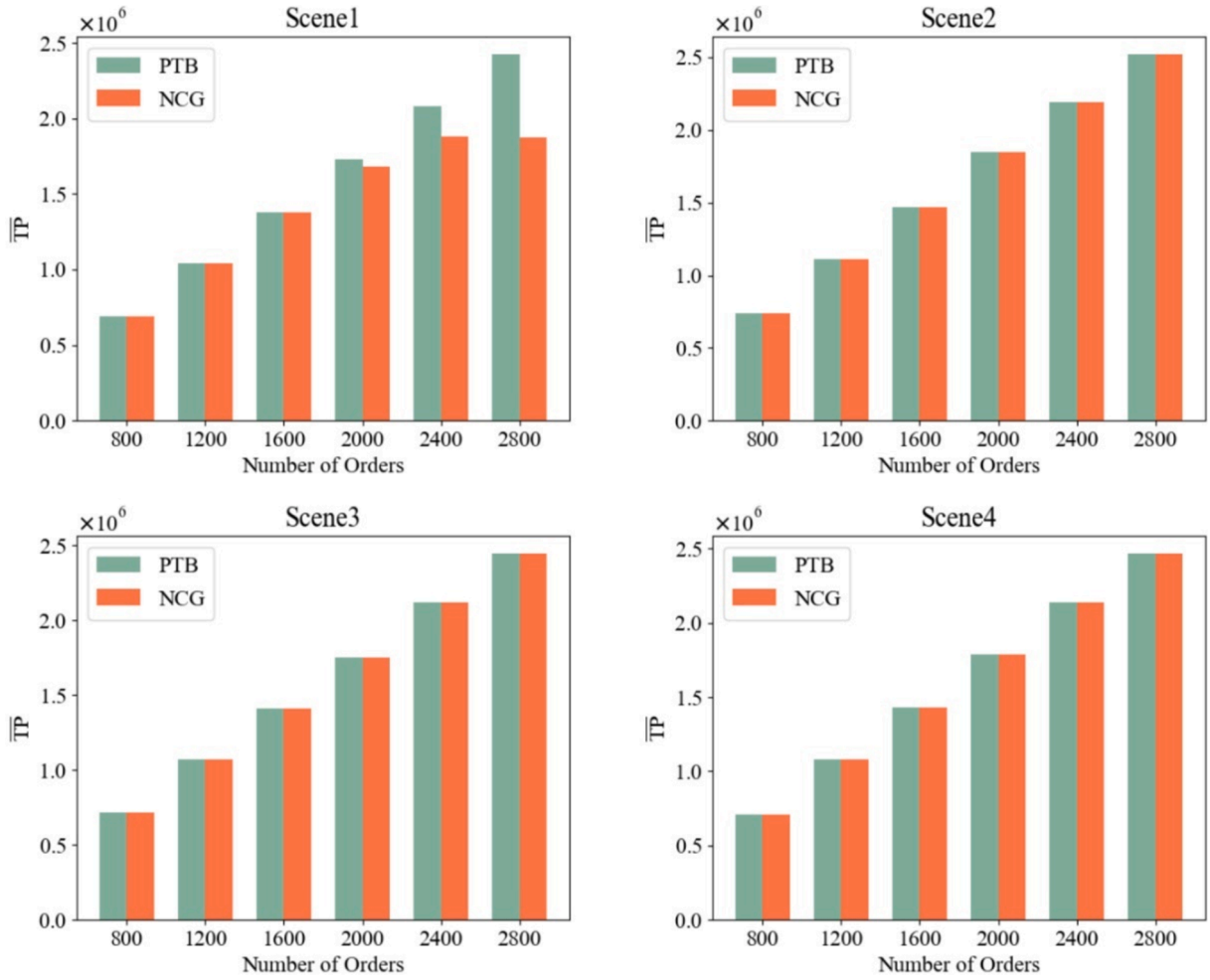


Fig. 7. Model profit comparison.

experimental results show that no matter in scene 1, scene 2, scene 3 or scene 4, the solving time of the PTB model for orders of different sizes is significantly lower than that of NCG algorithm. Meanwhile with the increase of order size, the gap between the solving time of the PTB model and NCG algorithm becomes larger and larger. This is mainly due to the parallel computing and sampling technology of the PTB model, which can quickly generate optimized decisions and thus show faster speed when solving problems, which makes the PTB model has obvious advantages when dealing with large-scale problems, and can give solutions in a short time. These results verify the rapidity and timeliness of the PTB model in solving problems, and prove the effectiveness and generalization ability of the PTB model in solving different order sizes from the side.

5.4.2. GVLS module performance verification

This section aims to examine the performance of the GVLS module when solving logistics order data with different numbers of service providers. When testing the performance of the model, the experiment uses the model weights trained based on 20 service providers to verify the order data containing 20 and 15 service providers; The model weights trained on 25 service providers were used to verify the order data of 20 and 25 service providers respectively. Further, the order data of 20, 25 and 30 service providers are validated using the model weights trained on 30 service providers. Finally, the order data for 20, 25, 30, and 35 service providers was validated based on the weights of the model trained by 35 service providers. The number of orders will remain the same in all scenes at 1000 to ensure consistency of experimental conditions. Finally, the Gap value between the verification results and the optimal value in each scene is calculated to evaluate the generalization ability and performance of the model. It should also be noted that when calculating the value of index TP and Gap, we adopt the calculation method defined by formula (72).

$$\text{Gap} = \left(\frac{|\text{TP}_{\text{OptimalValue}} - \text{TP}_{\text{CurrentValue}}|}{\text{TP}_{\text{OptimalValue}}} \right) \times 100\% \quad (72)$$

where $\text{TP}_{\text{OptimalValue}}$ represents the theoretical optimal value of indicator TP, and $\text{TP}_{\text{CurrentValue}}$ represents the current TP value calculated

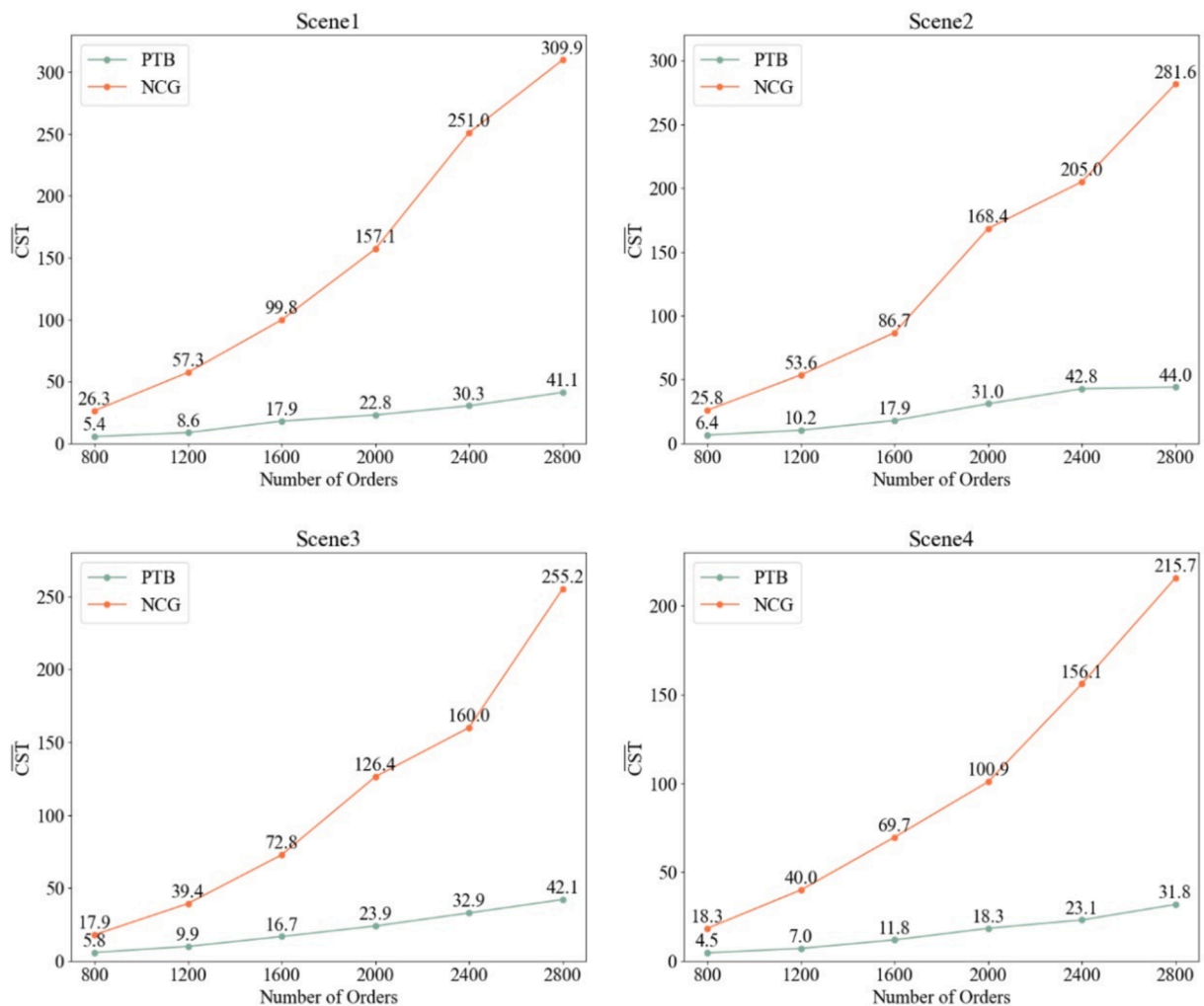


Fig. 8. Model time comparison.

by the PTB model. As the optimal value of STR and ETR is 0, AGap(Absolute Gap) is used to calculate the gap between the optimal value and the current value. The calculation method is shown in formula (73).

$$AGAP = |Indicator_{OptimalValue} - Indicator_{CurrentValue}| \quad (73)$$

where $Indicator_{OptimalValue}$ represents the theoretical optimal value of the indicator, and $Indicator_{CurrentValue}$ represents the current value

Table 6
GVLS result verification.

Train	Test	TP			STR			ETR		
		Opt	Ours	Gap	Opt	Ours	AGap	Opt	Ours	AGap
20	15	850975	428828	49.6 %	0 %	74 %	74 %	0 %	56.3 %	56.3 %
	20	870299	870299	0 %	0 %	2.7 %	2.7 %	0 %	0 %	0 %
25	20	870299	571196	34.3 %	0 %	63.5 %	63.5 %	0 %	40.9 %	40.9 %
	25	929783	929783	0 %	0 %	1.5 %	1.5 %	0 %	0 %	0 %
30	20	870299	527466	39.3 %	0 %	67.2 %	67.2 %	0 %	46 %	46 %
	25	929783	929783	0 %	0 %	2.9 %	2.9 %	0 %	0 %	0 %
35	30	889974	889974	0 %	0 %	0 %	0 %	0 %	0 %	0 %
	20	870299	833703	0.04 %	0 %	48.6 %	48.6 %	0 %	13.8 %	13.8 %
	25	929783	929783	0 %	0 %	0 %	0 %	0 %	0 %	0 %
	30	889974	889943	3.38 %	0 %	0 %	0 %	0 %	0.1 %	0.1 %
	35	904212	904212	0 %	0 %	0 %	0 %	0 %	0 %	0 %

of the indicator calculated by the PTB model. Finally, we summarized the verification results in Table 6 for detailed analysis and discussion.

It can be seen from the results in the Table 6 that the order data with different number of service providers in the four scenes all show good results in various evaluation indicators. For example, when the model weight of order data with 35 service providers is trained, the weight can not only solve the data size with 35 service providers, but also solve the data with 30,25, and 20 service providers, and the results are not weaker than those trained by them alone. The difference between the actual value and the optimal value is also smaller. This shows that the side-length sequence model with variable number of service providers is valid and has sufficient generalization.

From the data in the Table 6, it can also be found that when the model weight of 20 service providers is trained separately, it can perform excellent results in solving data with the same number of service providers. However, the model weights with 25, 30 and 35 service providers have the effect of decreasing transportation profit when solving the data with 20 service providers, especially when the number of service providers in the model weights is 25 and 30 respectively, the decrease of transportation profit is particularly serious. The reason for this is that the data set of 35 service providers is more complex than the data set of 20 service providers, so the

Table 7

Comparative study on ablation model (Bold is optimal, * is suboptimal).

Scene	Indicator	PTB-M1	PTB-M2	PTB-M3	PTB (ours)
Scene1	$\overline{TP}$	389012*	198807	367310	870299
	$\overline{TP}_{ES}$	123.72 %	337.76 %	136.94 %	—
	$\overline{STR}$	0.78*	0.89	0.79	0.03
	$\overline{STR}_{ES}$	−96.56 %	−96.99 %	−96.59 %	—
	$\overline{ETR}$	0.60 *	0.80	0.62	0.00
	$\overline{ETR}_{ES}$	−100.00 %	−100.00 %	−100.00 %	—
	$\overline{TTS}$	896*	1203	899	588
	$\overline{TTS}_{ES}$	−34.38 %	−51.12 %	−34.59 %	—
	$\overline{CST}$	10.94	13.81	9.87*	9.71
	$\overline{CST}_{ES}$	−11.24 %	−29.69 %	−1.62 %	—
Scene2	$\overline{TP}$	605522*	340092	368835	929783
	$\overline{TP}_{ES}$	53.55 %	173.39 %	152.09 %	—
	$\overline{STR}$	0.67*	0.83	0.79	0.01
	$\overline{STR}_{ES}$	−97.79 %	−98.20 %	−98.12 %	—
	$\overline{ETR}$	0.45*	0.70	0.65	0.00
	$\overline{ETR}_{ES}$	−100.00 %	−100.00 %	−100.00 %	—
	$\overline{TTS}$	824*	968	1016	611
	$\overline{TTS}_{ES}$	−25.85 %	−36.88 %	−39.86 %	—
	$\overline{CST}$	9.73*	11.65	11.58	7.32
	$\overline{CST}_{ES}$	−24.77 %	−37.17 %	−36.79 %	—
Scene3	$\overline{TP}$	889943*	488300	409074	889943
	$\overline{TP}_{ES}$	0.00 %	82.25 %	117.55 %	—
	$\overline{STR}$	0.00*	0.73	0.76	0.00
	$\overline{STR}_{ES}$	0.00 %	−99.86 %	−99.87 %	—
	$\overline{ETR}$	0.00*	0.53	0.60	0.00
	$\overline{ETR}_{ES}$	0.00 %	−99.81 %	−99.83 %	—
	$\overline{TTS}$	658*	940	1042	561
	$\overline{TTS}_{ES}$	−14.74 %	−40.32 %	−46.16 %	—
	$\overline{CST}$	8.85*	10.61*	12.31	5.41
	$\overline{CST}_{ES}$	−38.88 %	−49.01 %	−56.05 %	—
Scene4	$\overline{TP}$	904212*	672485	904212*	904212
	$\overline{TP}_{ES}$	0.00 %	34.46 %	0.00 %	—
	$\overline{STR}$	0.00*	0.59	0.00*	0.00
	$\overline{STR}_{ES}$	0.00 %	−100.00 %	0.00 %	—
	$\overline{ETR}$	0.00*	0.30	0.00*	0.00
	$\overline{ETR}_{ES}$	0.00 %	−100.00 %	0.00 %	—
	$\overline{TTS}$	442	777	370*	370
	$\overline{TTS}_{ES}$	−16.29 %	−52.38 %	0.00 %	—
	$\overline{CST}$	7.23	9.19	5.39*	4.57
	$\overline{CST}_{ES}$	−36.79 %	−50.27 %	−15.21 %	—
Total	Opt-Sub	0–17	0–1	0–6	20–0

* Notes: The statistical results of Opt and Sub are analyzed based on the data of indicators $\overline{TP}$, $\overline{TTS}$ and $\overline{CST}$, while the relevant results of indicators $\overline{TP}_{ES}$, $\overline{TTS}_{ES}$ and $\overline{CST}_{ES}$ are not included in the statistical scope of Opt and Sub. In order to ensure the accuracy of the effect size calculation formula, when the values of indicators $\overline{STR}$ and $\overline{ETR}$ for the comparative method are 0.00, then $\overline{STR}_{ES} = (\overline{STR} - \overline{STR}_{Comparative}) \times 100\%$, $\overline{ETR}_{ES} = (\overline{ETR} - \overline{ETR}_{Comparative}) \times 100\%$.

trained model may be too complex to accommodate the greater data complexity. When this complex model is applied to datasets with fewer service providers, it may inadvertently conform to data noise or minor features rather than the dominant patterns, which can lead to a degradation in performance.

5.5. Ablation study

5.5.1. Impact analysis of network module Reduction

In order to verify the effectiveness of modules in PTB, four variants of PTB are constructed in this section: (1) Replace Transformer module in policy network with GRU. (2) PTB-M2: Replace the Bayesian convolution and Bayesian linear layers in the final output layer in the policy network with ordinary convolution and ordinary linear layers. (3) PTB-M3: The convolutional layer in which decoder extracts information of different scales in the policy network is replaced with a linear layer.

The experiment trained data containing 1000 logistics orders in four scenes, and used the same metrics as the comparison experiment to evaluate the ablation model performance. As shown in Table 7, it can be observed that the performance of the PTB-M1 is worse than that of the PTB model. In scene 1, the effect size of indicators of the PTB model compared with that of the PTB-M1 reaches 123.72 %. This gap arises because Transformer can capture context information in sequences more comprehensively through its self-attention mechanism, while GRU can maintain state when processing long sequences, but its capture of context information is not as comprehensive as Transformer. The performance of the PTB-M2 is also inferior to that of the PTB model. In scene 1, the effect size of indicators of the PTB model is 337.76 % compared with that of the PTB-M2. This gap arises because Bayesian convolution and Bayesian linear layers can model uncertainties, which can then be better processed. But ordinary convolution and linear layers are unable to model the uncertainty, and thus lead to poor results in these cases. The performance of PTB-M3 is also inferior to that of the PTB model.

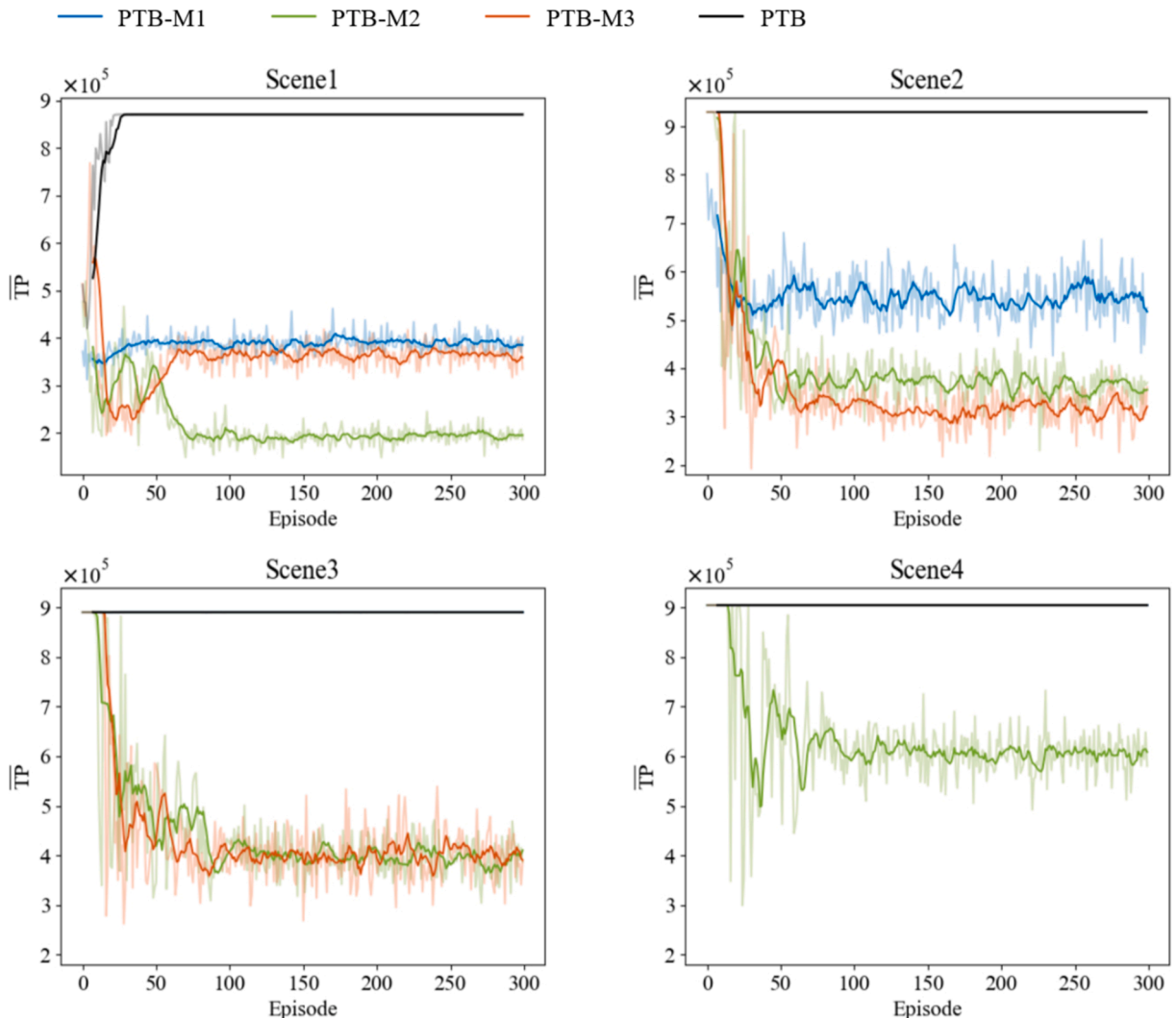


Fig. 9. Ablation model profit fitting curve.

In scene 1, the effect size of indicators of the PTB model reaches 136.94 % compared with that of the PTB-M3 model. The reason for this gap is that the convolutional layer can extract features such as details and textures in the image through local perception and multiple sliding operations of the convolutional kernel. However, in the absence of spatial awareness, the linear layer cannot capture these detailed features, resulting in reduced model representation. Finally, from the comparison results of various indicators, the PTB model is superior to other ablation models, which proves the effectiveness of the PTB model.

In Fig. 9, the horizontal axis represents the number of training iterations and the vertical axis represents the profit earned per iteration. According to the convergence curves of the PTB model and other ablation models in each scene, the convergence effect of the PTB model is better than that of other ablation models in most scenes. Meanwhile we can also see in Scene 4 that the convergence curves of PTB-M1 and PTB-M3 basically overlap with the PTB model, indicating that when the number of service providers is large, Replacing the convolutional layer used by decoder to extract information of different scales in policy network with linear layer or changing Transformer module to GRU has little impact on the model effect.

In the process of model training, the starting point value can reflect the fitting degree of the initial model to the objective function. It can be observed in Fig. 9 that the starting point of the PTB model is relatively high, that is, it is closer to the objective function than the other two ablation models at the beginning of training. Meanwhile it also shows that the PTB model is easier to move to the optimal solution faster because it needs to move less distance. For models with higher starting points, it is easier to converge to the optimal solution faster and get better results in a shorter time. Conversely, the other three ablation models may need to move farther during training to reach the location of the target data.

Meanwhile we can observe the unique advantages of a policy network that integrates Transformer and Bayesian networks in Fig. 9. Transformer can effectively capture the long-term dependencies of orders at different stages and optimize sequential decisions for service selection, while Bayesian networks introduce uncertainty modeling, allowing the model to make more robust predictions in the face of incomplete information. This combination not only improves the accuracy of decision making, but also adapts to the changeable logistics environment, so as to achieve the optimal service combination selection under the dynamic changing conditions. If changes are made to the Transformer structure or Bayesian network, the results can be significantly reduced to meet the FLSCPS

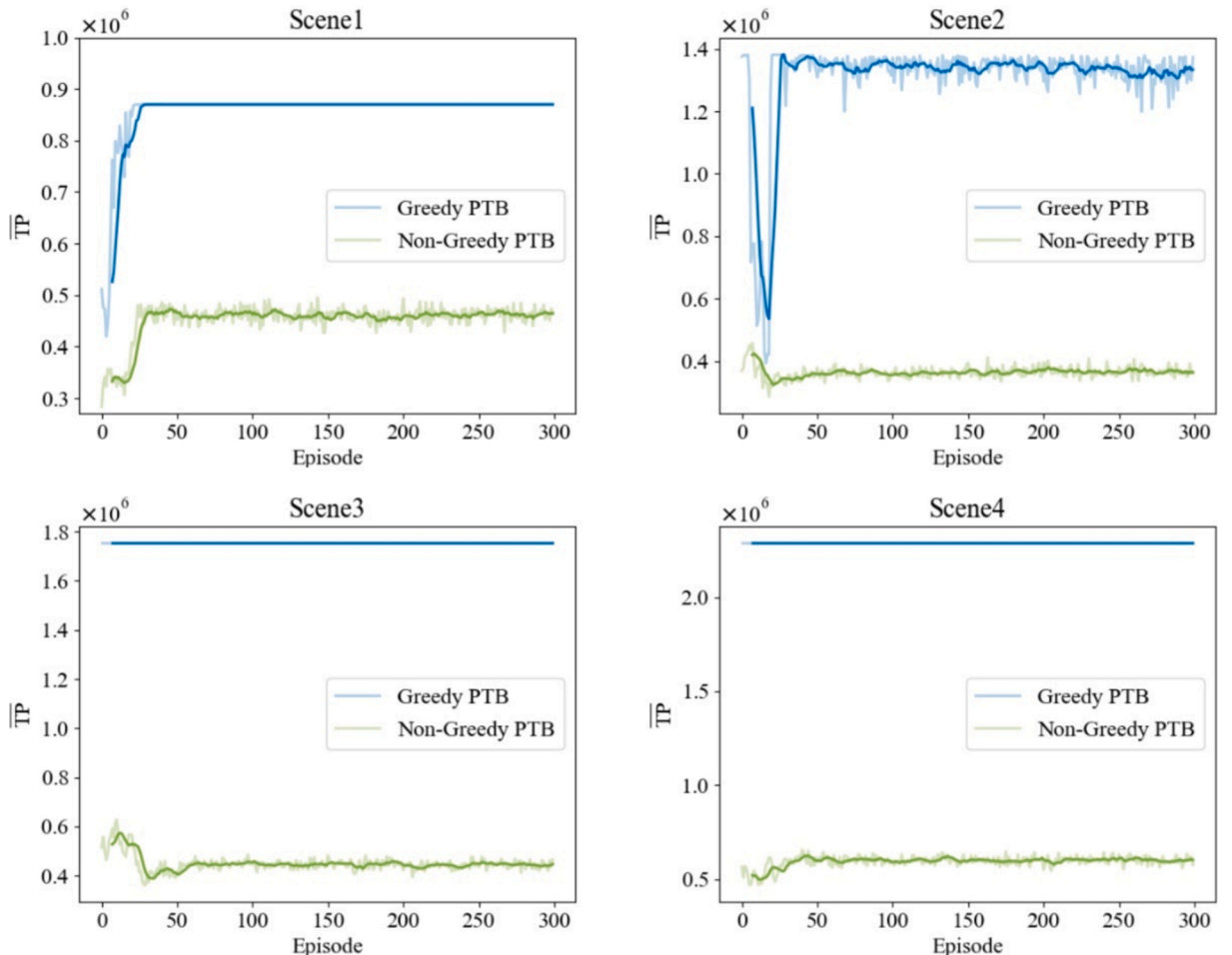


Fig. 10. Greedy strategy effect evaluation.

needs for fast and efficient solutions.

5.5.2. Evaluation of the effects of greedy strategy

In order to optimize the training efficiency of the PTB model, we propose a state information ranking method based greedy strategy. The method first selects a number of orders in the environment, and then selects the service providers that process these orders most efficiently. After the agent introduces actions to the environment, these service providers are responsible for optimizing the processing of the selected orders. With this approach, we try to improve the efficiency of model training and find the optimal strategy in the decision-making process.

To verify the effectiveness of greedy strategy in improving model training efficiency and selecting optimal strategies, we conducted experiments in four different training scenes. Each scene has its specific data Settings, where the maximum number of service providers for the first scene is 20 and the number of logistics orders available for transport is 1000. In the second scene, the maximum number of service providers is 25 and the number of logistics orders available for transport is 1500. In the third scene, the maximum number of service providers is 30 and the number of logistics orders available for transport is 2000. The maximum number of service providers in the fourth scene is 35 and the number of logistics orders available for transport is 2,500. These experiments are set up to provide a broader training environment to test the validity of our methods, independent of specific data Settings. Comparing the performance of the models in different scenes can more accurately evaluate the actual effect of greed in improving training efficiency and selecting optimal strategies.

Meanwhile in order to verify whether the Greedy strategy will affect the output strategy, we compare two situations that one is using greedy strategy, greedy-PTB, in the environment, and the other is not using greedy strategy, Non-Greedy-PTB, to observe the difference between them. In four different training scenes, we trained two models with and without greedy strategies and compared their convergence curves. The experimental results are displayed in Fig. 10, with the horizontal axis representing the number of training rounds and the vertical axis showing the overall profit earned from shipping.

By comparing the training effects of the model without the greedy strategy, we can intuitively see that in each training scene, the

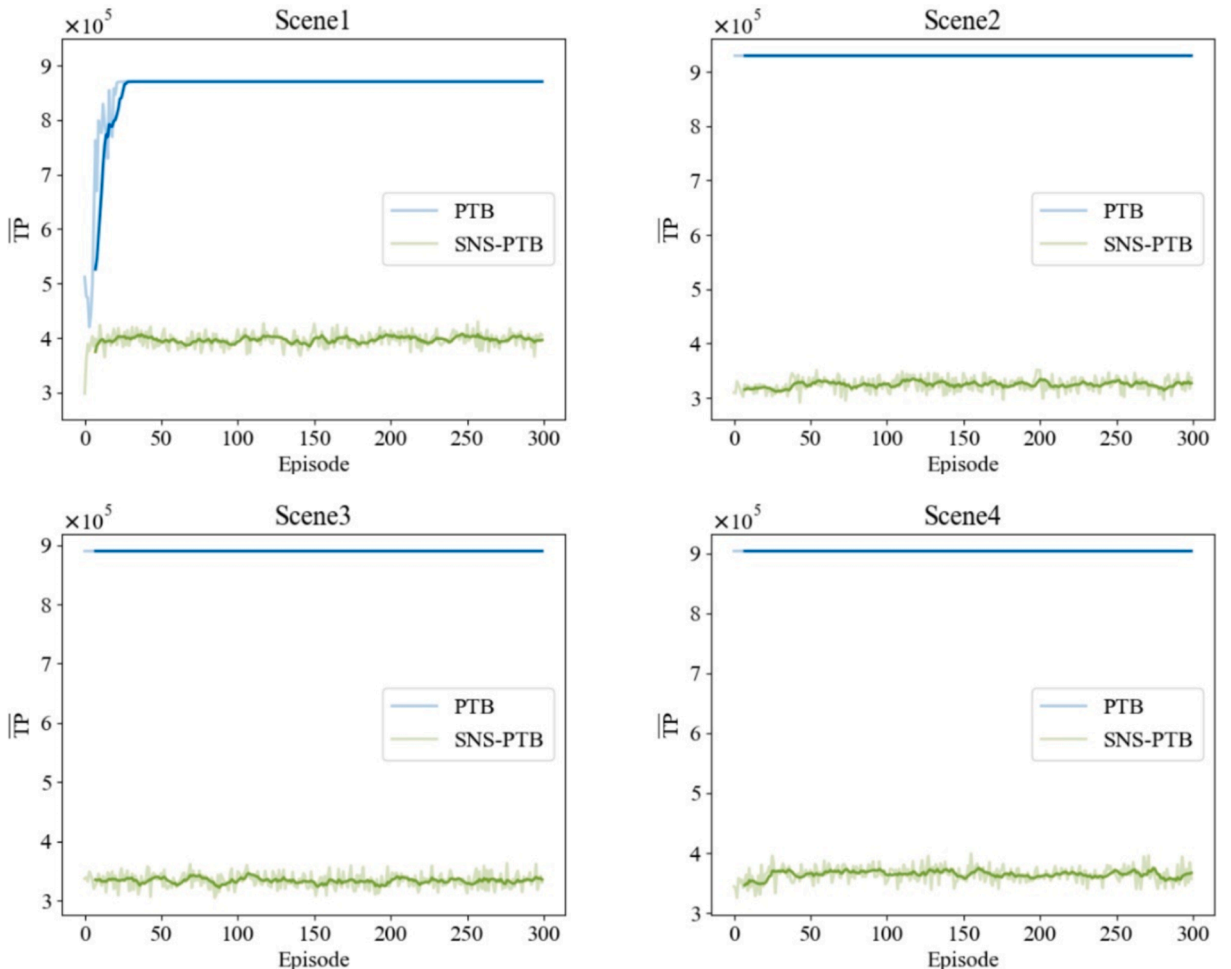


Fig. 11. Network structure difference assessment.

model using the greedy strategy shows a better trend in terms of the overall profit gained from transportation. This further proves the validity of greedy strategy in the process of optimization model training. After using this strategy, the training efficiency of the model and the selection of the optimal strategy have been significantly improved. The comparison results provide strong evidence that this optimization strategy is effective in improving the efficiency of model training and selecting the optimal strategy.

5.5.3. Policy vs value network structural differences

In the PTB model, the structure of policy network and value network is significantly different. The policy network consists of Encoder, Decoder and Bayesian module, while the value network consists of four layers of Bayesian convolution and two layers of Bayesian linear layer. In order to explore the effect of structural differences between policy network and value network on the model performance, we adjust the policy network to the same structure as the value network, and remove the greedy strategy to eliminate its interference on the new model structure performance. The newly established comparison model is called the PTB model of the same network structure (SNS-PTB). Then, in four scenes, we use 1000 logistics orders to train the PTB model and the SNS-PTB model respectively, and finally evaluate the impact of network structure differences on model performance by comparing the profit convergence of the two models.

Compare the results by observing at the models in Fig. 11, it can be found that the profit convergence of the PTB model is significantly better than that of SNS-PTB model in various scenes. This is mainly due to its unique strategic network structure, which is composed of Encoder, Decoder and Bayesian module. Meanwhile it draws on the advantages of the Transformer structure and adds sparse attention mechanism to realize effective extraction of state features. In addition, Bayesian modules are able to model the uncertainty of order data for more precise scheduling decisions. Greedy strategy also plays a key role in this model. It can selectively introduce the order states that are more in line with the processing capability of the service provider into the neural network, which provides a more reliable basis for the decision-making actions of the agent.

Meanwhile it can be observed that the structural differences between the policy network and the value network contribute to the improvement of the model performance. The policy network is mainly used to formulate the behavior strategy of the agent in a specific state, and its output is the action probability. Its structural design should fully consider the characteristics of different actions and adapt to environmental changes, so it usually includes deep network structure and complex models. In contrast, the task of a value network is

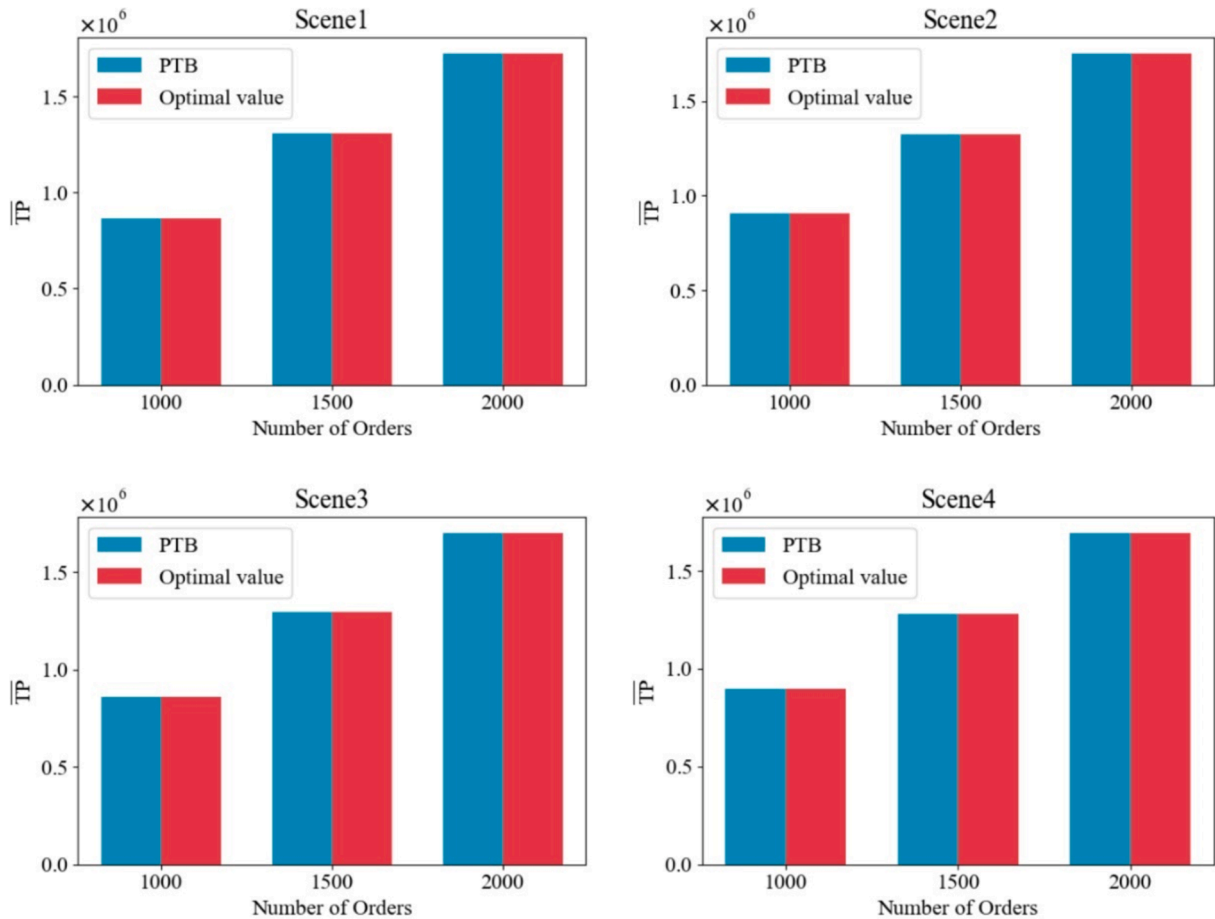


Fig. 12. Model robustness analysis.

different, and its primary goal is to evaluate the long-term value of a state or state-action pair. The structure design of the value network needs to be able to effectively learn the value information of the state without directly output the probability distribution, so its structure is more concise than that of the policy network.

5.6. Robustness study

The objective of a robustness experiment is to evaluate how the model performs when encountering anomalies or diverse distributions within the training data. Such tests can help to assess the robustness and generalization of the model, that is, how well the model performs in the face of unknown or noisy data. By testing data sets with different distribution of training data, we can better understand the actual application capabilities of the model and avoid unexpected situations in the production environment.

In the previous model training and testing phase, the logistics order data used in the FLSCPS environment met the uniform distribution. In order to verify the robustness of the model, we generate order data satisfying the Poisson distribution, and select all scenes as validation environments, with the number of order data being 1000, 1500 and 2000, respectively. The robustness of the model can be assessed by observing its performance on the newly distributed data set.

According to the experimental results shown in Fig. 12, the PTB model shows a remarkable effect on the data set conforming to the Poisson distribution. The model shows the results consistent with the optimal profit in each scene. For example, in scene 4, when the sample size is 1000, 1500 and 2000, the total profit of the model is 897161, 1,282,261 and 1,692,767 respectively, which is exactly consistent with the optimal value and presents zero error. This shows that the PTB model has strong robustness and can obtain stable and reliable results on different distributed data sets.

5.7. Parameter study

5.7.1. Sensitivity analysis of network depth

In order to further study the influence of hyperparameters on the performance of the PTB models, the experiments in this section

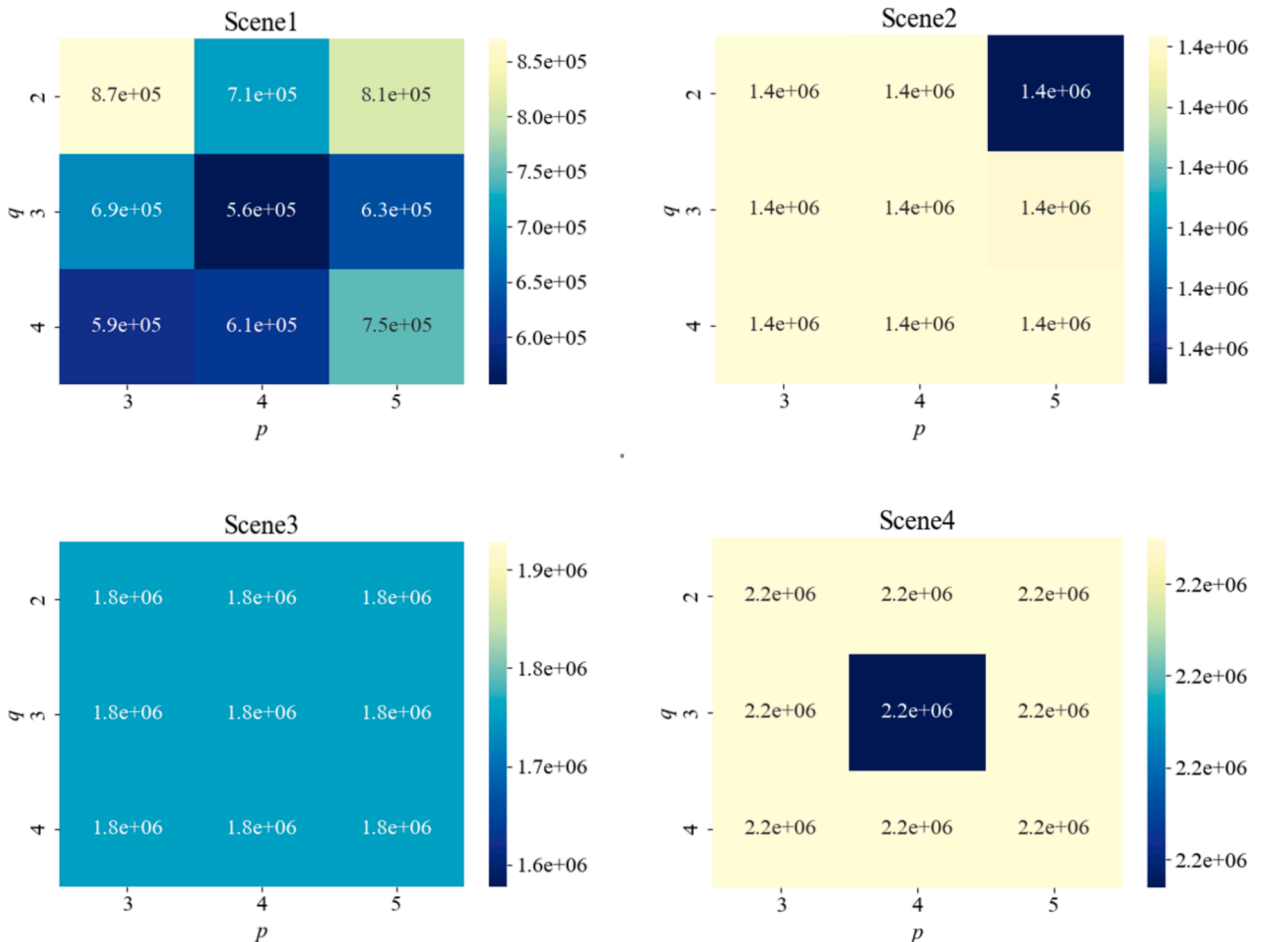


Fig. 13. Sensitivity analysis of Bayesian networks depth.

will focus on the number of Bayesian convolutional layers and the number of Bayesian linear layers in the Bayesian module of the policy network, and explore whether the ratio between them is related to the model performance.

In the experiment, the number of Bayesian convolutional layers $p = 3, 4, 5$, the number of Bayesian linear layers $q = 2, 3, 4$, and other parameters remain unchanged. Scene 1 contains 1000 logistics orders, Scene 2 contains 1500 logistics orders, Scene 3 contains 2000 logistics orders, and Scene 4 contains 2500 logistics orders. During the training process, the same number of training rounds and learning rate were maintained as in the ablation experiment. Fig. 13 shows a thermal map representation of the effect of p to q ratios on model performance. Among them, dark blue indicates the best effect and white indicates the worst effect.

In scene 1, when parameters $p = 3$ and $q = 2$, the overall profit of the model reaches a maximum. It can be observed that an increase in q value has a negative effect on model performance when p values remain constant. This is because an increase in the depth of the Bayesian linear layer leads to an increase in the number of model parameters, which increases the risk of overfitting. Especially in the case of limited training data, too many parameters will increase the complexity of the model and reduce its generalization ability. However, in scene 2, scene 3 and scene 4, we find that no matter what value p and q are taken, the influence on model convergence is very limited, and the best results can be achieved. This is because the number of service providers in scene 2, scene 3, and scene 4 is relatively large and the problem complexity is relatively small compared to scene 1. Therefore, increasing the depth of the network layer does not significantly affect the convergence effect of the model.

In some cases, a network that is too deep can cause overfitting problems, especially if the data is limited. It is important to choose the appropriate model architecture and parameter Settings to improve the generalization ability of the model and adapt to different scenes. Therefore, the policy network of the PTB model uses the network depth combination scheme of $p = 3$ and $q = 2$ to reduce the complexity of the network.

5.7.2. Time efficiency analysis of network depth

In order to study the effect of the depth of Bayesian network on the time efficiency of model training, the results of scene 1, in which the depth of network has the most significant effect on the model effect in the previous section, are selected for discussion. Meanwhile to facilitate graphical representation, we excluded the case where the number of convolutional layers is 5. Finally, the impact of network depth on the time efficiency of model training is assessed by comparing and analyzing the time required to reach optimal profit across different scenarios.

Six different network combination models were constructed by fixing the number of Bayesian convolution layers to 3 and 4 respectively and changing the number of Bayesian linear layers to 2, 3 and 4 respectively. Fig. 14 depicts the overall profit convergence of these models during the training process, where the horizontal axis represents the training time and the vertical axis represents the profit over time.

The results show that increasing the number of network layers does not significantly improve the convergence speed of the model. Specifically, the model achieves optimal performance with a configuration of 3 convolutional layers and 2 linear layers, reaching this peak after approximately 200 s. However, other models not only fail to surpass this configuration in training effect, but also fail to reach the optimal state stably during training. Therefore, we conclude that increasing the number of network layers does not always lead to better convergence, but may be accompanied by an increase in training time.

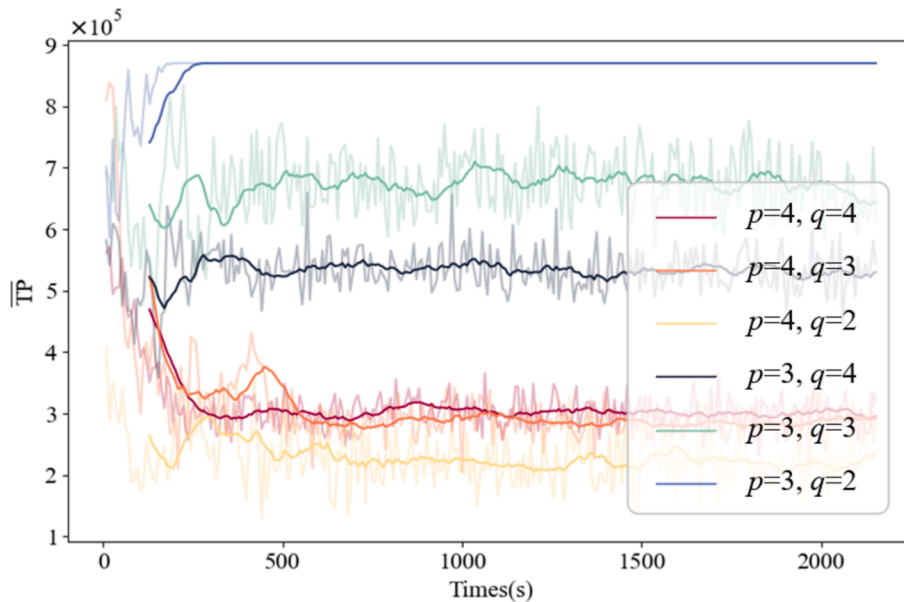


Fig. 14. Time efficiency analysis of Bayesian networks depth.

5.7.3. Learning rate parameter adjustment analysis

To investigate the specific impact of hyperparameters on the PTB model's performance, we conducted a thorough analysis of the policy network's learning rate while holding other parameters steady. In four different experimental scenes, we fixed the number of logistics orders at 1000 and adjusted the learning rate of the PTB model policy network to observe the intuitive impact of this factor on the model performance. In addition, we aim to determine an optimal strategic network learning rate to achieve the goal of obtaining the optimal logistics service portfolio solution. The experimental results will be visually presented in the form of a dot plot to more clearly reveal the intrinsic relationship between the learning rate and the model performance, with the horizontal axis representing the different learning rate values from $2e-8$ to $2e-3$, and the vertical axis representing the overall profit obtained by the PTB model in a specific scene.

Fig. 15 reveals the significant impact of learning rate on model performance through detailed comparative analysis of profit data corresponding to different learning rates in each scene. In scene 1, when the learning rate is $2e-4$ and $2e-3$, the PTB model achieves the highest total profit, which is 870299 , respectively, indicating that in this scene, the higher learning rate is conducive to improving the economic benefit of the model. However, in scene 2, the situation is different. When the learning rate is $2e-8$, the profit obtained by the model (913500) is significantly better than that obtained by other learning rates, which suggests that a lower learning rate is more appropriate in this scene. Further, in the $c = 30$ and $c = 35$ scenarios, it is observed that the profit distribution presents a certain stability and diversity. For scene 3, when the learning rate is $2e-8$, $2e-7$, $2e-6$, and $2e-3$, the model can maintain a high profit level (889974), while in scenario 4, the learning rate except $2e-5$ ($2e-8$, $2e-7$, $2e-6$, $2e-4$, $2e-3$) can achieve an optimal profit (904212). This phenomenon shows that in these two scenarios, the model's sensitivity to learning rate is relatively low, and a variety of learning rate settings can achieve ideal performance.

Based on the above analysis, we find that the choice of learning rate has a critical impact on the overall profit of the PTB model, and the impact varies with the number of service providers. In certain scenes, a lower learning rate (e.g. $2e-8$) can effectively avoid overfitting, thus improving the economic benefits of the model. In other scenarios, appropriately increasing the learning rate (e.g. $2e-4$, $2e-3$) can help speed up model convergence and maximize profits. Therefore, in practical application, the learning rate should be flexibly adjusted according to the specific number of service providers and business needs to optimize the comprehensive performance of the PTB model.

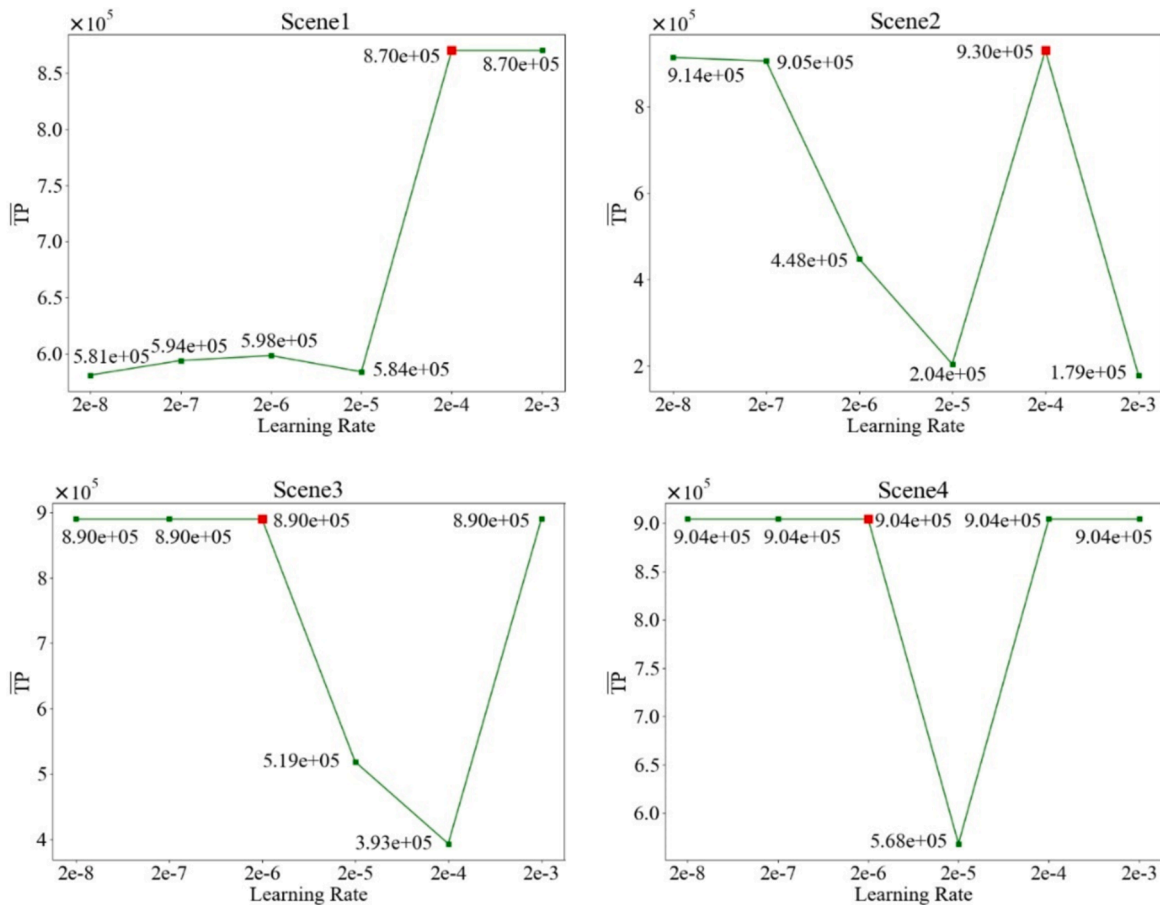


Fig. 15. Learning rate performance impact analysis.

5.7.4. Epoch parameter adjustment analysis

In order to further investigate the specific effect of hyperparameters on the performance of the PTB model, the epoch meter values of PPO algorithm are studied in detail, while other parameters are kept constant. In four different experimental scenes, we fixed the quantity of logistics orders as 1000, and adjusted the epoch meter value of PPO algorithm to 10, 15, 20, 25, and 30, respectively. In addition, the optimal learning rate in [section 5.7.3](#) was used in each experimental scene. Finally, the effect of epoch on the performance of the model is observed through the change of the total profit obtained by the model.

As can be seen from the data in [Table 8](#), as the epoch meter value of PPO algorithm increases, the performance of the model in different scenes shows different trends. In scene1, when the epoch count is increased from 10 to 15, the overall profit obtained by the model increases, but after that the overall profit remains the same, at 870299, no matter how the epoch is increased. This indicates that in this particular scene, the epoch count reaches 15, and the model has already converged, and increasing the epoch further will not provide any performance improvement. In scene2, as the epoch meter value increases from 10 to 25, the total profit made by the model first increases and then decreases, reaching a peak of 920,206 when the epoch is 25. This indicates that in this scene, an appropriate epoch count can improve the model performance, but an excessively high epoch value may degrade the model performance. In scene3 and scene4, no matter how the epoch meter values change, the total profit made by the model remains the same, 889,974 and 904212, respectively. This shows that the epoch meter value has no significant effect on model performance in either scenario.

Based on the above analysis, we can conclude that in the PTB model, the impact of the epoch meter value on the model performance varies from scenario to scenario. In some scenarios, an appropriate epoch meter value can improve the performance of the model, but in other scenarios, increasing the epoch meter value has little or no effect on the performance of the model. Therefore, in actual applications, you need to adjust the epoch meter to find the optimal hyperparameter configuration.

5.8. Model performance verification under large-scale orders

To verify the performance of PTB model when solving large-scale logistics orders, we do further research. In various scenarios, 5000, 7500 and 10,000 orders were selected as examples of model testing to observe whether the PTB model can still obtain higher transportation profits under large-scale order data. Meanwhile the model's transportation profit under varying order scales can be observed as it changes with an increase in the number of service providers. The objective is to identify the minimum number of service providers required to handle a fixed number of orders while maximizing profit, thereby providing valuable reference for the logistics industry.

In the validation process against large-scale order data, it can also be observed that the PTB model solution continues to perform well on large-scale order data. Regardless of the number of orders, the PTB model is capable of achieving the optimal overall transportation profit when the number of service providers is 25, 30, and 35. When the number of service providers is below 25, the PTB model achieves lower transportation profits. This is mainly due to the relatively small number of service providers, which makes the number of orders that each service provider can process in a timely manner limited when processing large-scale order data, resulting in more heavy tasks and thus less efficient processing.

In [Fig. 16](#), we can see that when the order size is 5000 and 7500, the PTB model also significantly increases the shipping profit when the number of service providers increases. Especially when the number of service providers is 22, the model's transportation profit has basically reached peak. When the order size increases to 10000, the transportation profit reaches the maximum when the number of service providers is 23. The data reveals that the minimum number of service providers should be 22 in order to process 5000 and 7500 order data and obtain a high profit, while the minimum number of service providers should be 23 in order to process 10,000 order data and obtain maximum profit. Therefore, when optimizing transportation solutions for large-scale order data, multiple factors such as the number of service providers, task volume and service efficiency need to be taken into account. These findings help to improve and optimize logistics and supply chain systems, thereby increasing efficiency and profitability.

Overall, the PTB model still shows high processing efficiency when processing large-scale order data. This is mainly due to its powerful parallel processing capabilities, which can effectively cope with fluctuations in task loads and maintain high processing speed, and these observations further confirm the advantages and potential of PTB models in processing large-scale order data.

6. Managerial insights

The PTB model as an important tool of modern logistics management, with its unique algorithm design and efficient data processing ability, has brought a full range of management advantages for enterprises. The following will describe the significant benefits of the PTB model at the management level from four aspects: decision efficiency, cost control, team collaboration and decision quality.

Table 8
Impact of Epoch on model performance.

Scene	Epoch				
	10	15	20	25	30
Scene1	870299	420187	870299	870299	870299
Scene2	913500	906548	878958	920206	867518
Scene3	889974	889974	889974	889974	889974
Scene4	904212	904212	904212	904212	904212

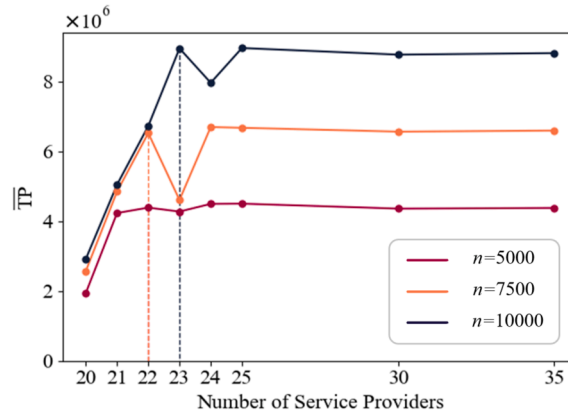


Fig. 16. Large-scale order model performance verification.

Firstly, the PTB model integrates the advantages of Transformer network and Bayesian network, which can analyze and process logistics data in real time to provide efficient decision support for managers. The application of this model reflects the modern practice of Taylor's scientific management theory, that is, to optimize work flow and improve work efficiency through scientific methods. In logistics management, this means that you can quickly respond to market changes and adjust service plans in time to meet customer needs. For example, when a service provider is unable to provide services in an unexpected situation, the PTB model can quickly generate alternative shipping solutions to ensure that orders are delivered on time. In addition, the model can optimize resource allocation, reduce empty vehicles and idle storage space, and improve the speed and efficiency of logistics operations. Such optimization not only improves customer satisfaction, but also enhances the market response speed and competitiveness of enterprises.

Secondly, the application of the PTB model significantly reduces the non-essential expenses in the logistics process, which is in line with the core idea of cost management theory, that is, to achieve cost savings through refined cost control and efficiency improvement. In the transportation sector, the PTB model reduces transportation costs by optimizing the service combination plans, reducing mileage and fuel consumption. Meanwhile the model can optimize cargo loading and distribution plans, reduce losses and delays in transportation, and reduce the resulting fines and compensation costs. These cost savings are directly reflected in the financial position of the enterprise and provide financial support for the long-term development of the enterprise.

Thirdly, the implementation of the PTB model requires the close cooperation of multiple departments such as logistics, IT and marketing to participate in the design, implementation and optimization of the model. This mode of cross-departmental collaboration promotes information sharing and knowledge exchange, which conforms to the principle of synergy in team management theory. Through effective team collaboration, companies can respond quickly to market changes and solve problems efficiently, thereby improving overall operational efficiency and customer satisfaction.

Finally, the PTB model utilizes deep learning and reinforcement learning techniques to provide managers with data-driven decision support, which is a further development of Simon's bounded rationality model. In the case of incomplete information, PTB model can use the available information to help managers make closer to the optimal decision. The comprehensive data perspective provided by PTB model not only improves the accuracy of decision making, but also enhances the enterprise's ability to adapt to market changes

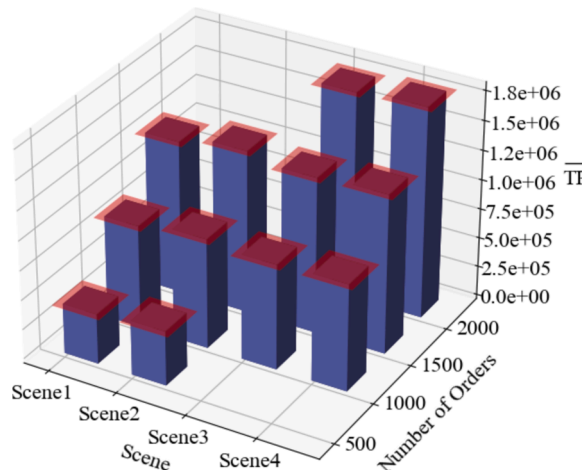


Fig. 17. The gap between total profits and optimal profits in various scenarios for the PTB model.

Table 9
Order service composition scheme.

Number	Collection	Trunk	Distribution	Start time	End time	Total time	Total profit
1	C_8	C_2	NULL	9	51	42	1875
2	C_4	NULL	NULL	9	10	1	79
3	NULL	NULL	C_3	1	9	8	252
4	C_{15}	NULL	C_4	39	128	89	207
...	...	...	...	...	...	...	...
401	C_1	NULL	NULL	207	208	1	69
402	C_5	NULL	C_{14}	279	320	42	328
403	C_{15}	NULL	C_{15}	288	299	11	367
...	...	...	...	...	...	...	...
1000	C_{15}	NULL	C_{13}	753	776	23	258

*Notes: NULL indicates that the order has no such service.

and control risks. Through continuous learning and optimization, the model continuously improves the performance of the decision network and provides reliable data support for the long-term development and strategic planning of enterprises.

In the actual application process, the PTB model optimizes the logistics service combination scheme through continuous trial-and-error interaction between Agent and FLSCPS environment. When the reward converges to a stable value, the training ends, and the optimal logistics service combination scheme is output. Fig. 17 details the difference between the total profit achieved by the PTB model and the optimal profit of the corresponding instance under four different scenarios. The blue bar chart represents the overall profit achieved by the PTB model, while the red plane represents the theoretical optimal profit for the instance.

Fig. 17 clearly shows that the PTB model achieves exactly the same result as its theoretical optimal profit in practical application. This shows that PTB model is not only effective in theory, but also shows high efficiency and accuracy in practice, and can make full use of given conditions to achieve the maximum profit goal. Such high-quality results show that the design and implementation of the model are successful, and can provide reliable data support for decision makers.

Table 9 shows part of the order service combination scheme after the training of the PTB model. Each row of data reveals how various service requirements in a specific order are allocated to different service providers, and includes information such as the start processing time, end processing time, total processing time and total profit obtained.

By observing at the data in Table 9, we can conclude that the PTB model can effectively generate the order service composition scheme in the data testing phase. Each line of data details how the requirements for each service in a particular order are intelligently distributed to different service providers, while providing critical time nodes and total profit information for order processing. This shows that the PTB model can not only reasonably plan service allocation to ensure timely processing of orders, but also optimize profit capture. The data in Table 9 shows the balance ability of the model among processing time, efficiency and service quality, which reflects that the PTB model has high practicability and economic benefits in practical application, and provides scientific and reasonable decision support for order processing.

7. Conclusion

In this paper, we deal with a flexible logistics service combination problem with spatial-temporal constraints, the difficulties are that service providers have service area and service time constraints when transporting logistics orders, customer service needs have uncertainty, and the network cannot solve the problem of having different amounts of service provider data. To address these challenges, we propose a PTB model that incorporates both Transformer and Bayesian networks. The model employs the near-end policy optimization (PPO) algorithm training the agent. The policy network is reorganized by Transformer architecture. The encoder converts the order state into a vector representing its characteristics, and the decoder integrates the output of the encoder with sequence order data with different scale information for processing and analysis. Finally, the prediction results of the model are obtained through the Bayesian network module, and more accurate scheduling actions are generated. Meanwhile we adopted the variable length sequence module GVLS based on greedy strategy in the environment, thus solving the logistics service mix problem of flexible service providers. In order to verify the effectiveness of the model, the PTB model is compared with other heuristic algorithms and DRL methods. The experimental results on four groups of logistics order scenes show that PTB is superior to the traditional heuristic algorithm and other reference models in terms of optimization quality and training efficiency. Meanwhile generalization experiment, ablation experiment, parameter experiment, large-scale data performance verification experiment and management insight are also carried out, which further proves the validity and practicability of the model.

Although the PTB model has demonstrated its excellent performance in several experiments, during the design process, we also noted that it may face challenges in application. Firstly, in hyperscale logistics environments, the computational and scaling capabilities of the PTB models can be limited, especially when it comes to processing large-scale data, performing parallel computing tasks, and meeting real-time response requirements. To address these challenges, we plan to explore efficient computing architectures and optimization strategies. We will focus on the following aspects: using distributed computing system and edge computing technology to enhance the parallel processing capability of the model; Use model pruning and quantification techniques to improve performance and reduce resource consumption; Meanwhile more efficient algorithms are studied to reduce the computational complexity and speed up the processing. Secondly, in the process of model development, we made simplified assumptions about the behavioral stability of

service providers and the statistical characteristics of order arrivals, which reduces the complexity of the problem, but may also affect the applicability of the model in different scenes. To this end, we will take the following measures: integrate dynamic factors into the scheduling environment of the model to improve its ability to adapt to the dynamic changes in reality; Analyze the model's performance in different application scenes, identify and adjust assumptions that limit the model's ability to generalize; Use data enhancement and transfer learning techniques to enrich training datasets and enhance the generalization ability of models in diverse environments. In short, we will continue to explore solutions to enhance the potential of the PTB model in hyperscale logistics environments, ensuring that it can provide accurate and efficient scheduling solutions to adapt to complex and variable application scenes.

CRedit authorship contribution statement

Ran Tian: Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Conceptualization. **Longlong Chang:** Writing – review & editing, Writing – original draft, Validation, Conceptualization. **Zhihui Sun:** Writing – original draft, Validation. **Guanglu Zhao:** Writing – original draft, Validation. **Xin Lu:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 1. . Notation

Table 1
Mathematical symbols and corresponding meanings in the model.

Symbols	Description
n	Total logistics orders
m	Total service providers
l	Service quantity of order o_i
T	Total time
t	Current system time
N_{co}	Processing completed orders
mc	The maximum number of service providers in a scenario
C	Collection of service providers
O	Order collection
D_i	Set of service requirements for order o_i
P_i	The profit set of all service requirements for order o_i
G	The total profit made by the service provider in the problem scene
A	Scheduling action set
R	Reward set
S	State space
cs	Whether there is a collection service
ts	Whether there is trunk service
ds	Whether there is delivery service
$No_{c_k}^t$	Number of services assigned to service provider c_k at time step t
De_{c_k}	A collection of services that service provider c_k can process
$Ip(c_k, d_i^j)$	Whether service provider c_k can handle the j -th service requirement d_i^j of Order o_i
$Bool_{cs}^t$	Time constraints for collection services at time t
$Bool_{ts}^t$	Time constraints for trunk services at time t
$Bool_{ds}^t$	Time constraints for delivery services at time t
Vd_t	Whether the service vehicle departs at time t
cw_t	Load of the service vehicle at time t
Lw_{max}	Maximum load bearing capacity of the service vehicle

(continued on next page)

Table 1 (continued)

Symbols	Description
V^{o_i}	Volume of order o_i
$Cons_{time}$	Service time constraints
$Cons_{space}$	Spatial constraints of services
$t_{previous}$	the last time point
$R_{resource}$	Resource state
$O^i_{attribute}$	static properties of order o_i
$OS_{o_i}(t)$	The status of order o_i at time t
$Co^t_{o_i}$	Whether order o_i processes completed Boolean variables at time t
$Op(t)$	The order processing progress at time t
$Nco(t)$	The number of completed orders that have been processed at time t
$Nop_{o_i}(t)$	The number of selections for order o_i at time t
$\Delta Nop_{o_i}(t)$	The number of new picks for order o_i during the interval from $t_{previous}$ to t
$Nosp_{o_i}(t)$	The optimal number of service providers assigned for order o_i at time t
Oc_t	Whether there is a Boolean variable of order completion in the time t environment
S_{static}	Static information in the state space
$S_{dynamic}$	Dynamic information in the state space
$Osst_{o_i}$	Whether the start processing time of order o_i is timed out
$Oest_{o_i}$	Whether the end processing time of order o_i is timed out
$T^{generate}_{o_i}$	The generation time of order o_i
$T^{sp}_{o_i}$	The start processing time of order o_i
$T^{res}_{o_i}$	Response time for order o_i
$T^{deadline}_{o_i}$	The deadline for order o_i
$T^{ep}_{o_i}$	The end processing time of order o_i
$T^{latest}_{o_i}$	The latest processing completion time for order o_i
$T^{timeout}_{d_j}$	Order o_i j -th service demand shipping timeout time
$T^{end}_{cs_{o_i}}$	The end time of the collection service for order o_i
$T^{begin}_{ts_{o_i}}$	The begin time of the trunk service for order o_i
$T^{end}_{ts_{o_i}}$	The end time of the trunk service for order o_i
$T^{begin}_{ds_{o_i}}$	The begin time of the delivery service for order o_i
TP	Indicator: total profit obtained
TTS	Indicator: the time it takes to process all orders
CST	Indicator: the solving time of the instance
STR	Order start processing timeout rate
ETR	Order end processing timeout rate
α	Profit attenuation coefficient
α_θ	Learning rate of policy network
α_v	Learning rate of value network

Data availability

Data will be made available on request.

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